

Weeks 2-10

K-1

Spoken instructions; matching, building, and small counts.

Week 2 is one shared collection; grade bands below apply to Weeks 3-10.

Give one page at a time. The packet is a menu, not a checklist for one hour.

Each student set includes the same Week 2 library. Print its chosen pages once.

Week and investigation	PDF pages
02 Switches and lamps	2-43
03 Repeating secret machines	44-46
04 Around the ring	47-49
05 Tower cities	50-52
06 Code-breaking with guarantees	53-55
07 Take-away games	56-58
08 Rooks and piles	59-61
09 Bouncing paths	62-64
10 Bridges and delivery routes	65-67

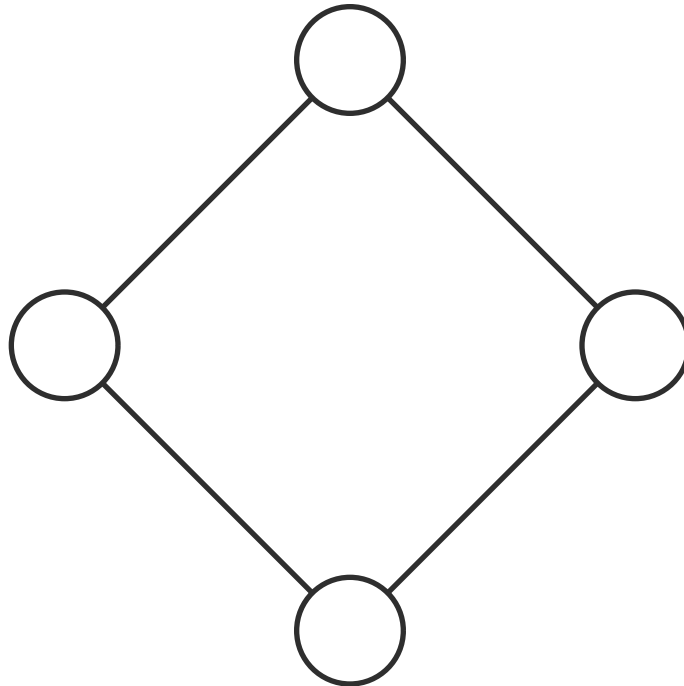
Print US Letter, single-sided, at 100% / Actual Size.

Page ranges above count this cover as page 1. Bookmarks open each week.

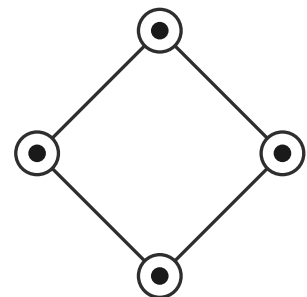
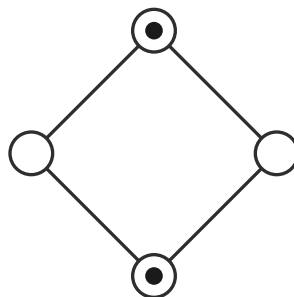
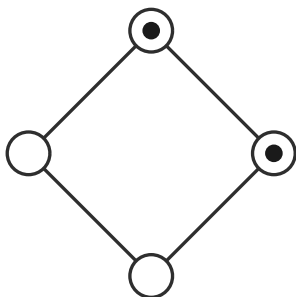
Weekly footers retain their own page numbers and activity IDs for the use log.

Unpiloted. Week 2 updated September 28; other core revisions September 27.

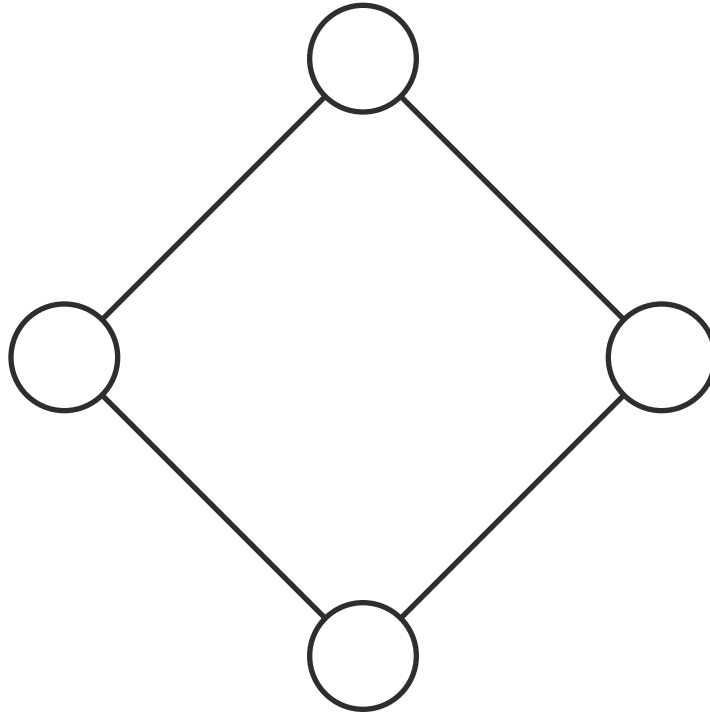
Problem 1: Empty is OFF. A dot is ON. Choose a line. Change BOTH lamps at its ends: add a dot or erase a dot. Try a move. Try that same move again.



Problem 2: Start all OFF each time. Make each small picture on the big board.

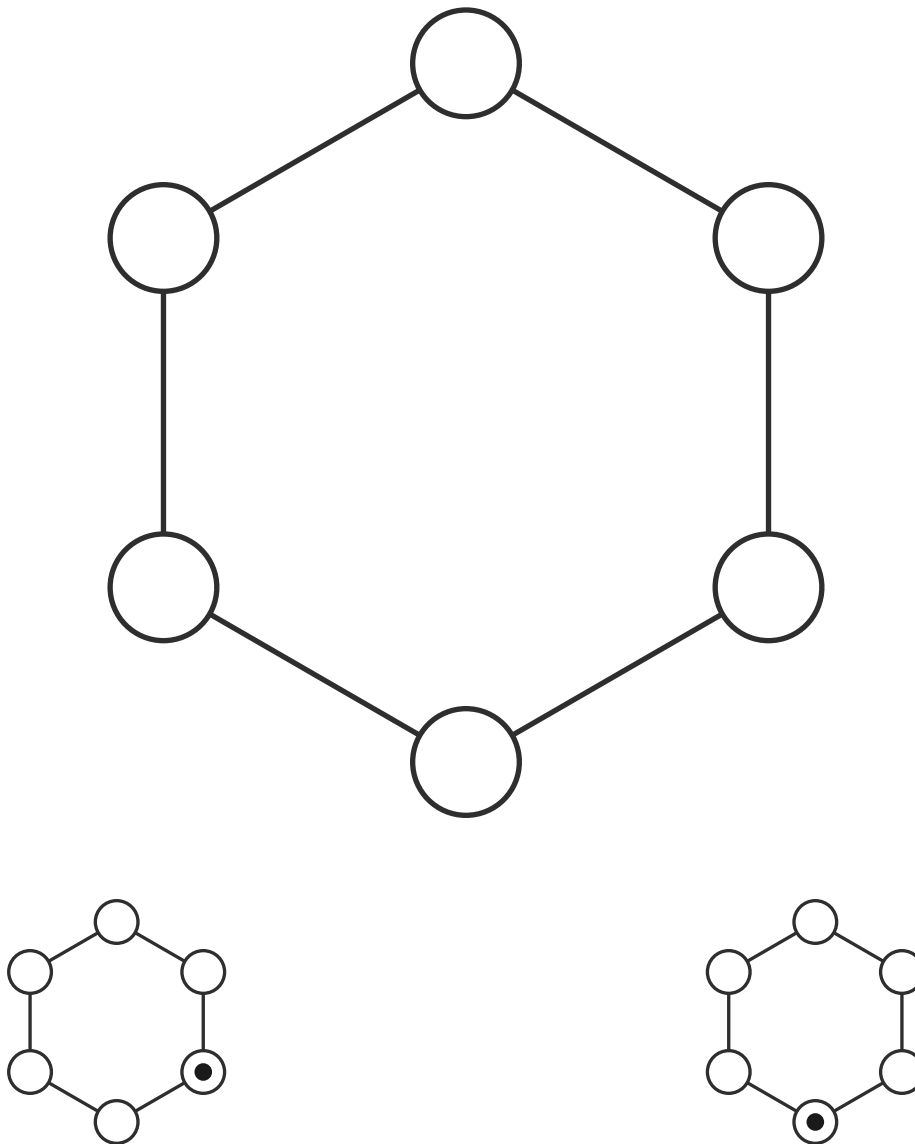


Problem 3: Play with a partner. Start all OFF. Make one move: change both ends of a line. Can your partner turn all the lamps OFF? Swap jobs.



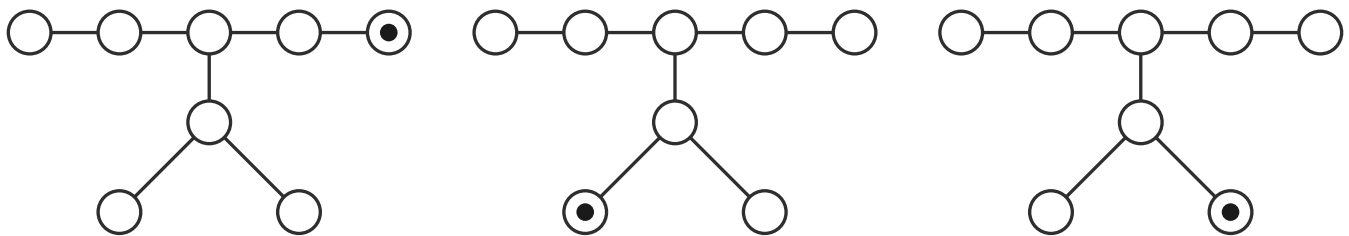
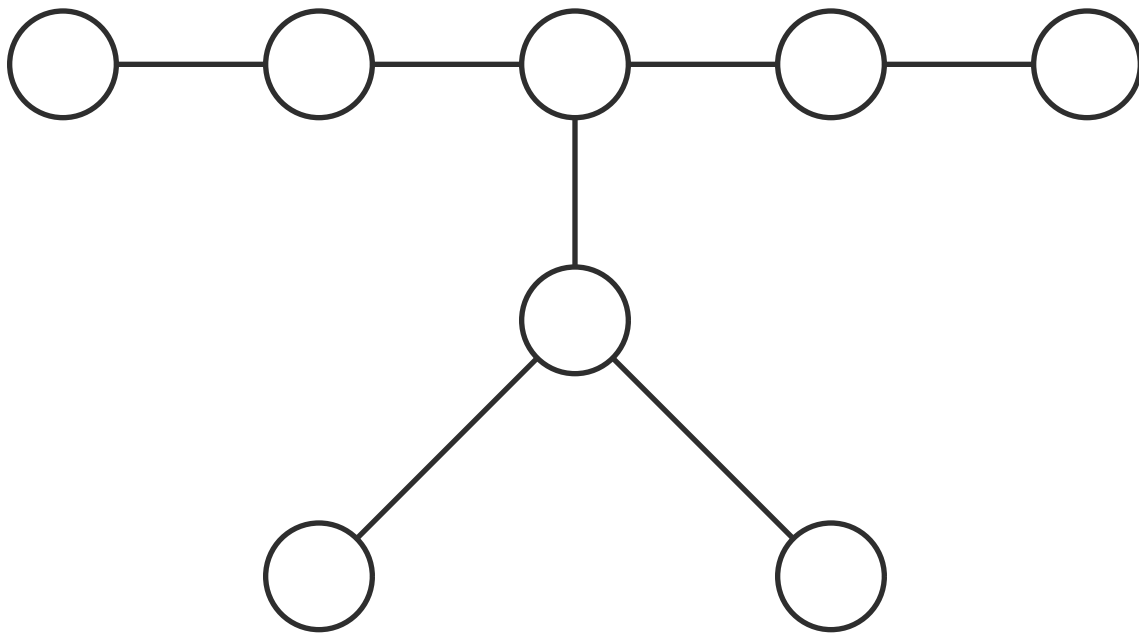
Problem 4: Start all OFF. Make TWO moves for your partner to undo. Swap jobs. Then try THREE moves. Show your moves if your partner wants a hint.

Problem 5: Start with only the top lamp ON each time. Change both ends of a line to make each small picture. Keep exactly ONE lamp ON after every move.

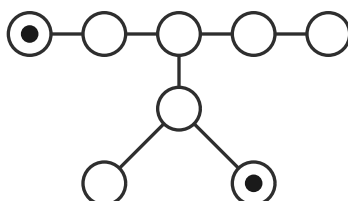


Problem 6: Start at the top again. Visit every lamp and bring the light home.

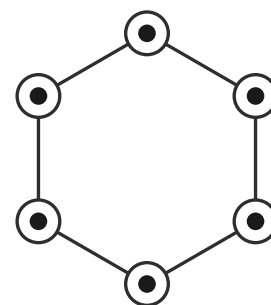
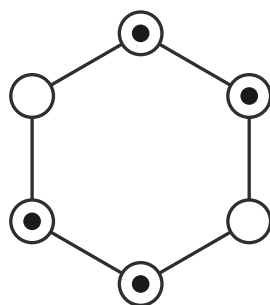
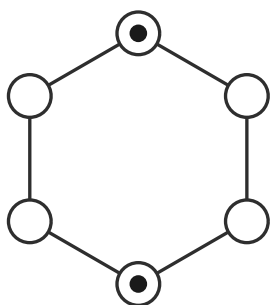
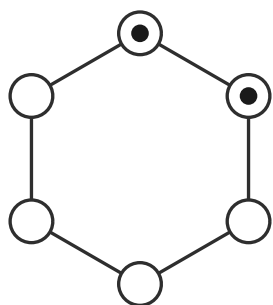
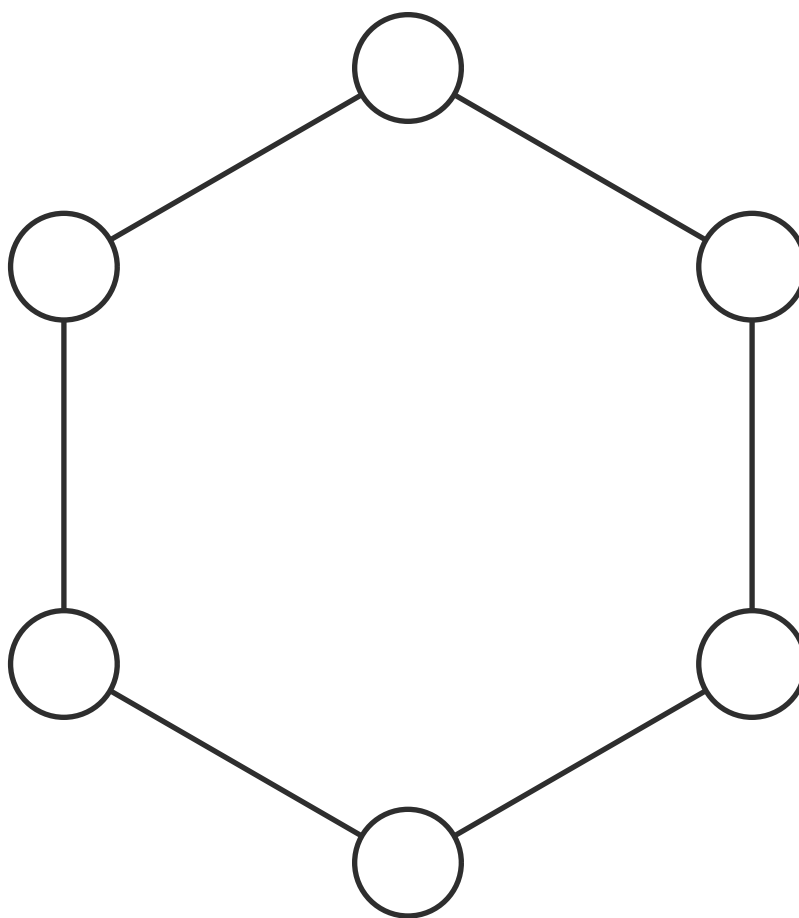
Problem 7: Start with only the far-left lamp ON each time. Change both ends of a line. Move the light to match each small picture. Keep ONE lamp ON.



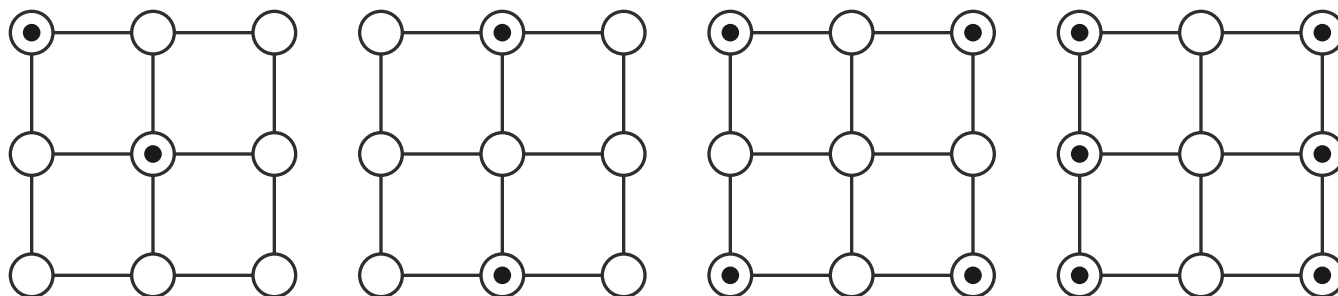
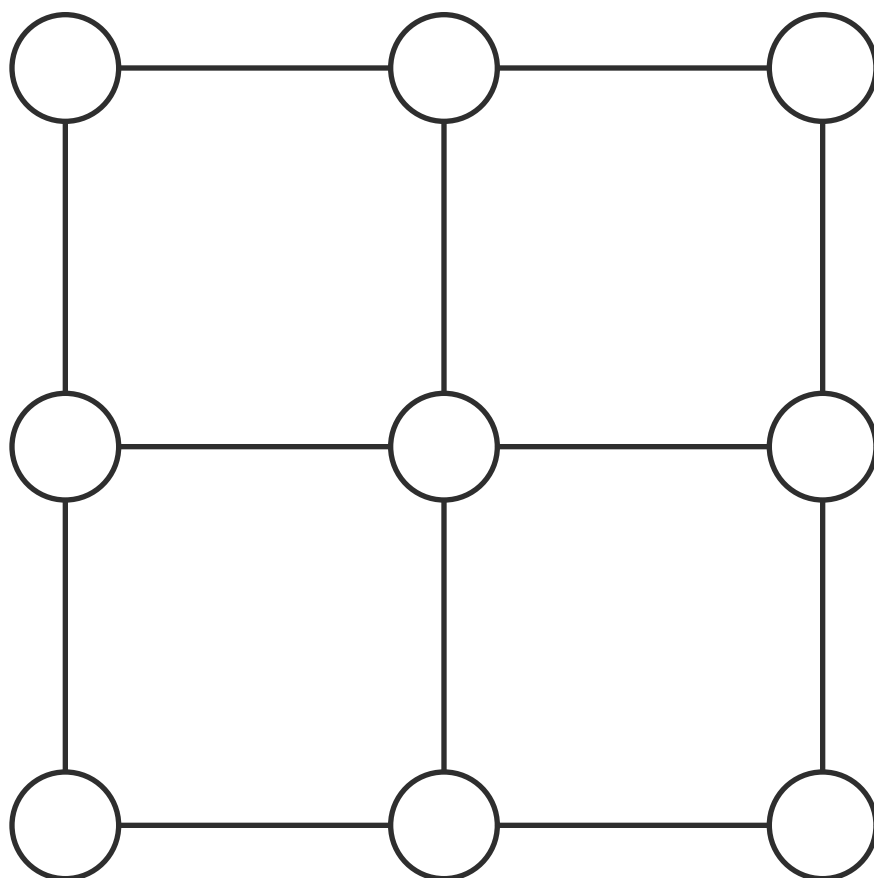
Problem 8: Now start all OFF. Make this two-light picture. You may have more than one light ON.



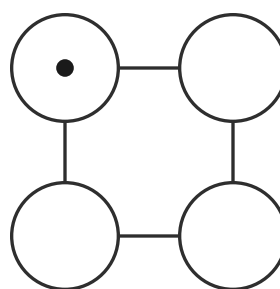
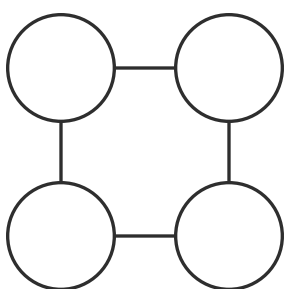
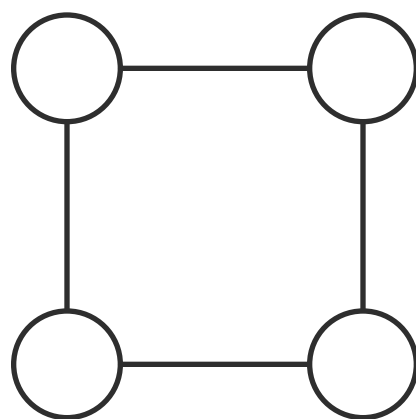
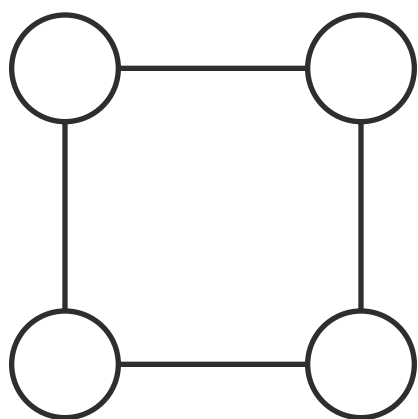
Problem 9: Start all OFF for each small picture. Change BOTH ends of one line each move. Make the picture on the big board. Point to one you want to do again.



Problem 10: Start all OFF for each small picture. Change BOTH ends of one line each move. Make the picture on the big board. Try a different way to make one picture.



Problem 11: Put a dot in the top-left lamp of the LEFT island. Change BOTH ends of one line each move. Can you make the small picture?



Problem 12: Draw ONE new line between lamps on different islands. Start again with only the top-left lamp of the left island ON. Try the trip with your bridge.

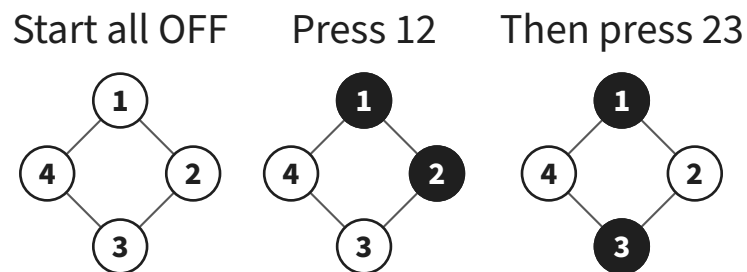
Problem 13: Draw 4 to 8 lamps. Join them with lines so every lamp has a path to every other lamp. Keep lines from crossing. Start all OFF. Make TWO moves, changing both ends of a line each time. Let a partner turn them all OFF.



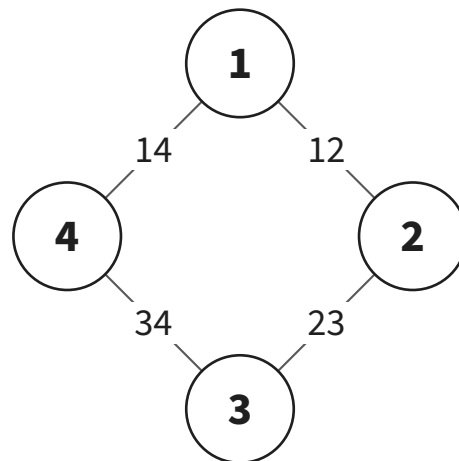
Problem 14: Add one new lamp. Join it to an old lamp with a line. Make a new puzzle for your partner.

Problem 15: Use the line names to save your moves. Start all OFF on the big board. Press a line: change *both* end lamps. Add or erase pencil dots, or flip two-sided counters. Dotted or dark means ON.

The line joining lamps 1 and 2 is called **12**. Replay these two moves on the big board.



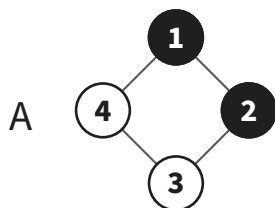
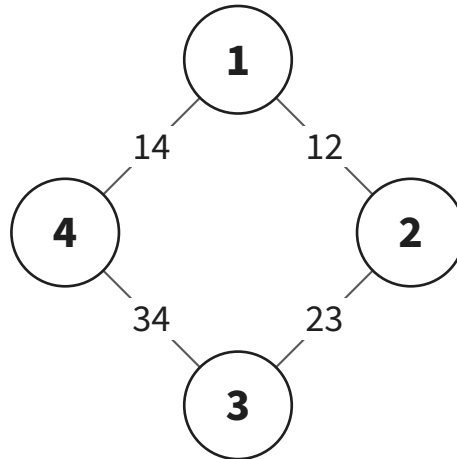
Save the moves in order: **12, 23**. You may ask someone to write while you point.



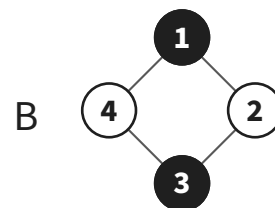
Get back to all OFF by pressing lines. Save your moves so a partner can replay them.

Moves: _____

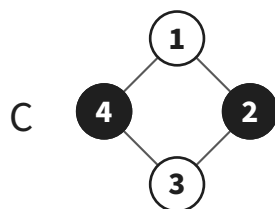
Problem 16: Make each target picture on the big board. Reset all lamps to OFF before each target. Change both ends of one line at a time. Write your moves in order.



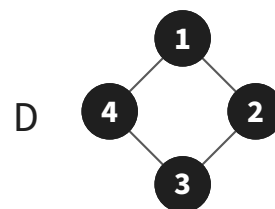
Moves: _____



Moves: _____



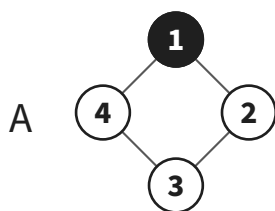
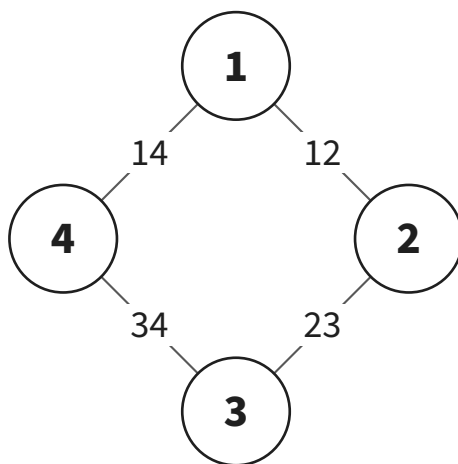
Moves: _____



Moves: _____

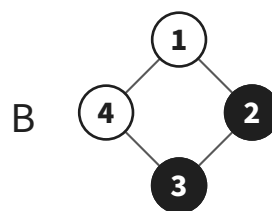
Give a partner one of your move lists. Reset all OFF. Have your partner replay it and point to its target.

Problem 17: Try each target picture. Start all OFF each time. Change both ends of one line at a time. Write a move list that works, or check “not yet.”



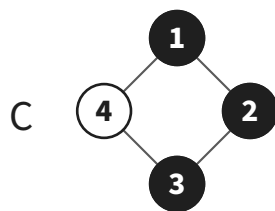
Moves: _____

☐ Not yet



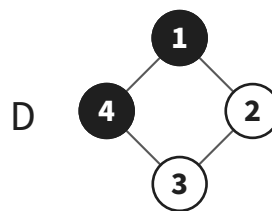
Moves: _____

☐ Not yet



Moves: _____

☐ Not yet

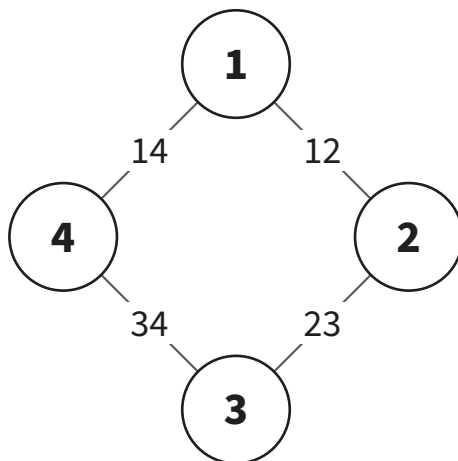


Moves: _____

☐ Not yet

For a “not yet” picture, try a different move list on the board. Keep any list that works.

Problem 18: For each row, set up its starting picture by hand on the big board: add dots or turn counters over. Leave lamps 3 and 4 OFF. Then press 12 once. Darken the ON lamps in the answer picture. Count ON lamps before and after.



A  — 12 — 
ON before: _____

press 12 →

 — 12 — 
ON after: _____

B  — 12 — 
ON before: _____

press 12 →

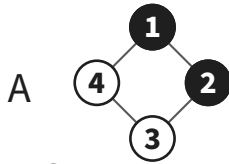
 — 12 — 
ON after: _____

C  — 12 — 
ON before: _____

press 12 →

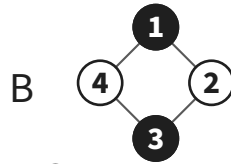
 — 12 — 
ON after: _____

Problem 19: These are the targets you tried. For each picture, circle “made” or “not yet.” Count its ON lamps. Use the big board to try any picture again; reset all OFF each time.



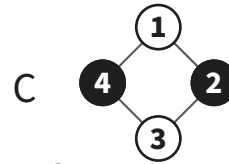
ON: _____

made / not yet



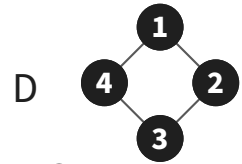
ON: _____

made / not yet



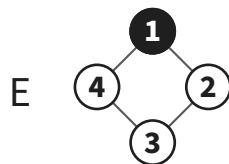
ON: _____

made / not yet



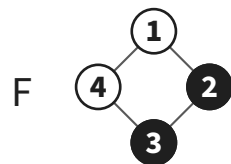
ON: _____

made / not yet



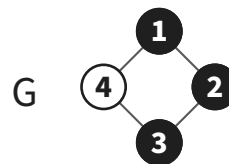
ON: _____

made / not yet



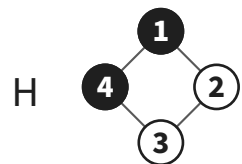
ON: _____

made / not yet



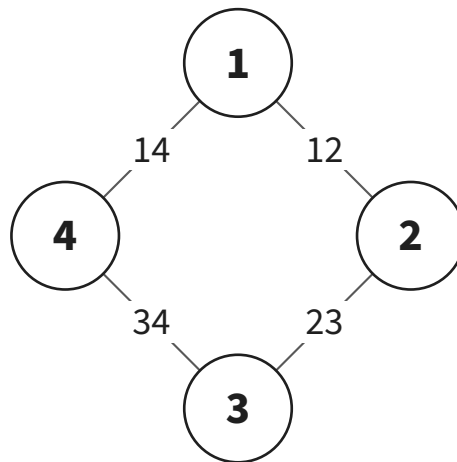
ON: _____

made / not yet



ON: _____

made / not yet

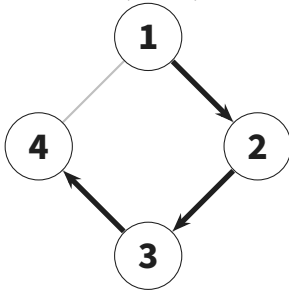


Compare the ON counts in the pictures you made.

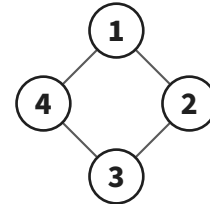
Point to what is alike. Then compare the “not yet” pictures.

Problem 20: Follow each arrow path on the big board: change both ends of each line, in order. Start all OFF for each path. Darken the ON lamps in its final picture.

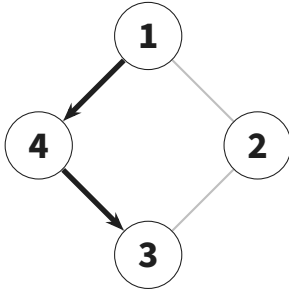
A: 12, 23, 34



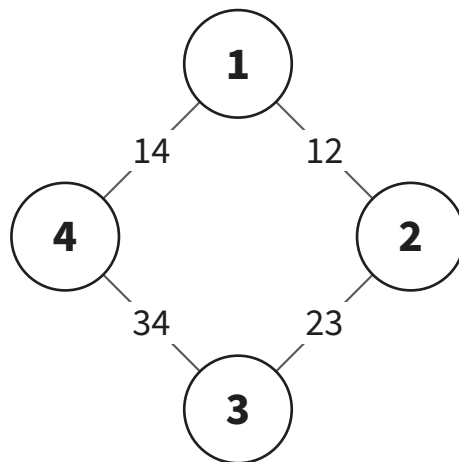
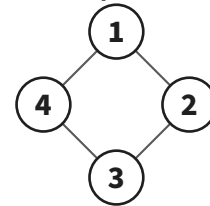
Final picture:



B: 14, 43

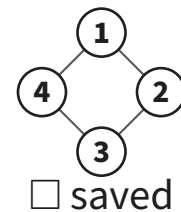
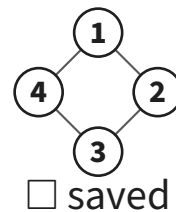
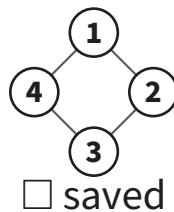
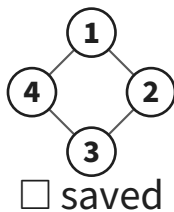
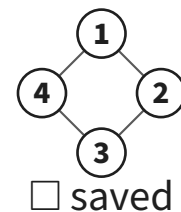
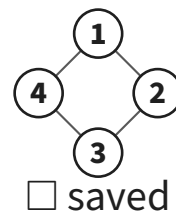
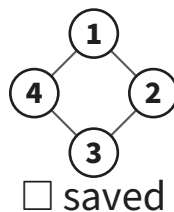
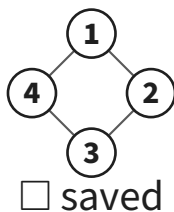
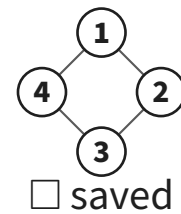
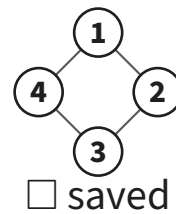
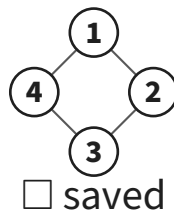
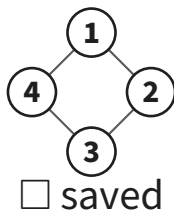
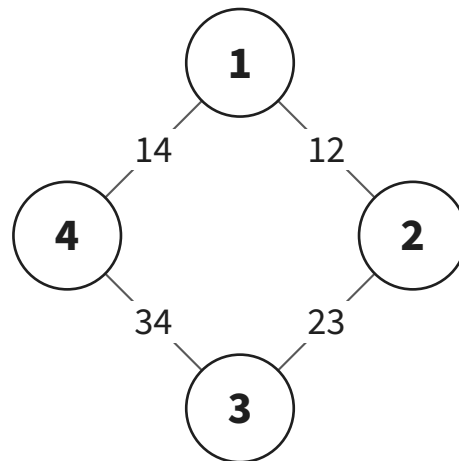


Final picture:



Replay one path slowly. Point to each lamp that changes twice. Point to each lamp changed just once.

Problem 21: Start all OFF on the big board. Try some moves. Copy the result onto a little ring: darken its ON lamps. Reset and make different pictures. Include all OFF. Leave extra rings empty.



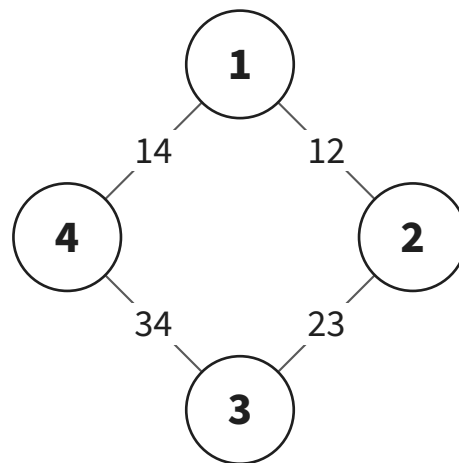
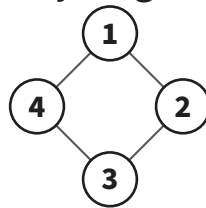
Check “saved” under each picture you record, including all OFF. This keeps unused rings separate.

Problem 22: Look at the pictures you saved in Problem 21. Count the ON lamps in each. Write that count beside its little ring. Point to the pictures with the same count. Count how many different pictures you have.

I saved _____ different pictures.

Choose a picture whose moves you have not written down. Copy it here. Start all OFF on the big board and find a move list that makes it.

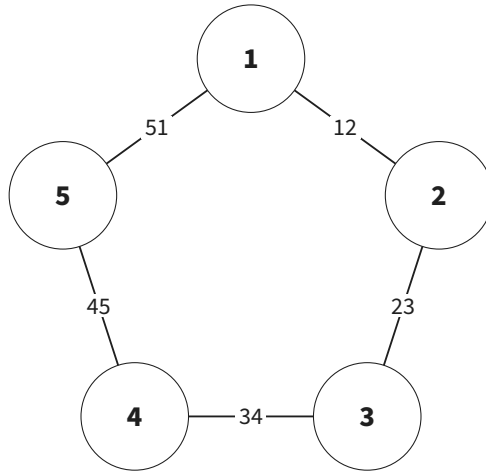
My target:



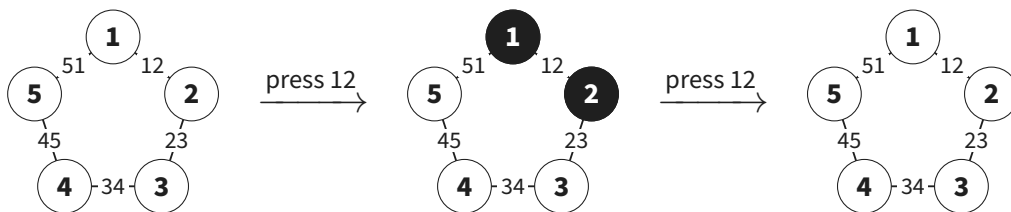
My moves: _____

Reset all OFF and ask a partner to replay your list. Compare the result with your target.

Problem 23: Try a five-lamp ring. Start all OFF on the large board. Press a line: change both end lamps. Add or erase pencil dots, or flip two-sided counters. OFF becomes ON; ON becomes OFF.



Write **12** for the line joining lamps 1 and 2. Dotted or dark means ON; white means OFF. You may ask someone to write your moves while you point.



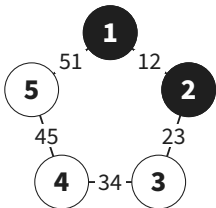
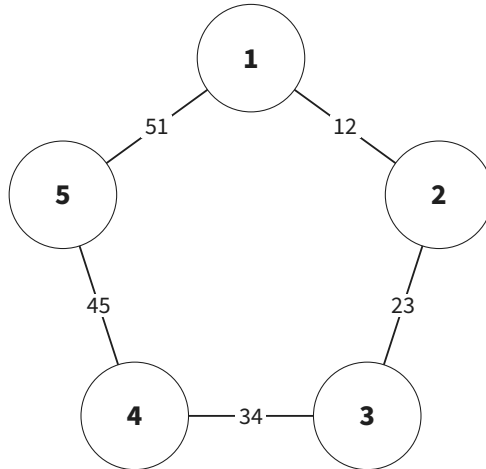
Try the two presses shown above. Then start all OFF and choose two presses of your own. Write them in the order you press them.

First press: _____

Second press: _____

Get back to all OFF. Write the presses you used.

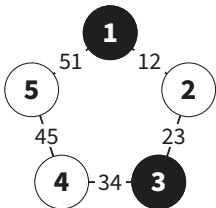
Problem 24: Start all OFF for each target below. Make the dark lamps ON and the white lamps OFF. Write two different move lists for each target. Circle the shorter list, or write “same” if they tie.



Make this picture.

First list: _____

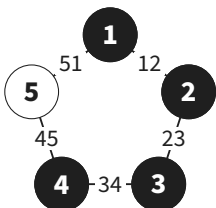
Another list: _____



Make this picture.

First list: _____

Another list: _____



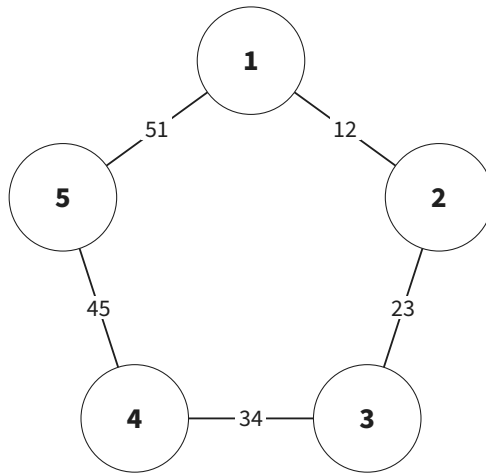
Make this picture.

First list: _____

Another list: _____

Choose one target. Test single presses, resetting to all OFF before each press. Can one press make that target? Tell a partner what happened.

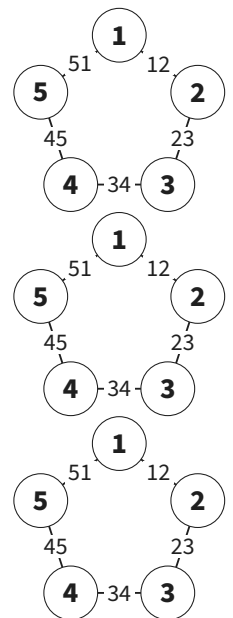
Problem 25: Start all OFF for each move list. Read from left to right. Press one line at a time, then color the lamps that finish ON in its small ring.



Press **12**, then **23**.

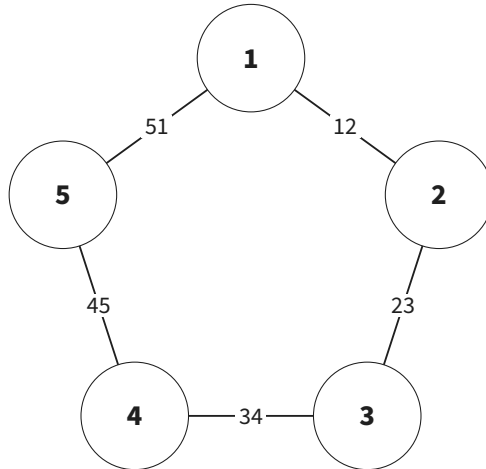
Press **23**, then **12**.

Press **12**, then **23**, then **12**, then **12**.



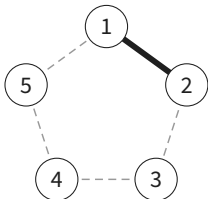
Compare the three pictures. In the last list, cross out two copies of the same press. Reset and try the presses left over. Do you get the same picture?

Problem 26: For each pair, start all OFF. Press each thick line on the left once, then color the lamps that finish ON. Reset to all OFF. Test the thick lines on the right and color that final picture.



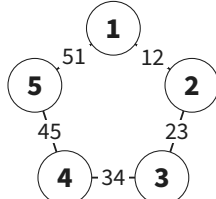
The dashed lines are the lines you do not press. In the first pair, try line 12 on the left; on the right try 23, 34, 45, 51.

Press these

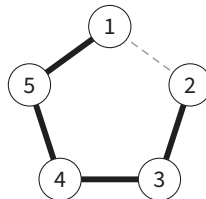


___ presses

Color result

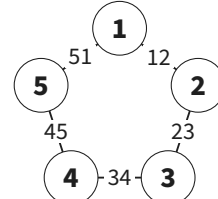


Reset. Press these

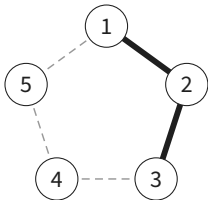


___ presses

Color result

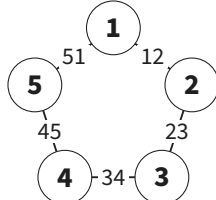


Press these

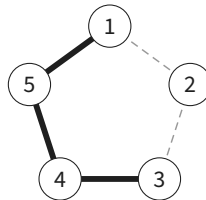


___ presses

Color result

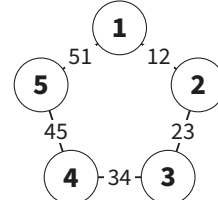


Reset. Press these



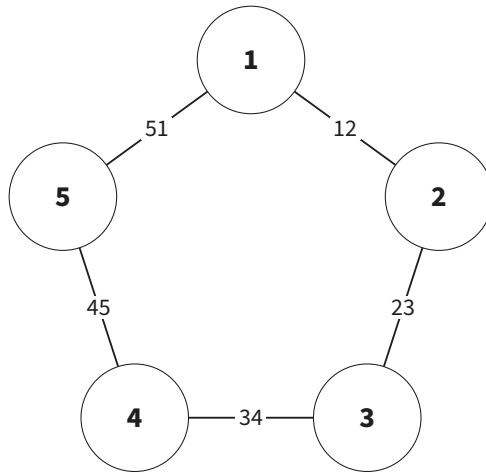
___ presses

Color result

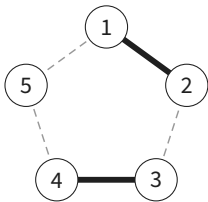


In each pair, count the presses and compare the two final pictures. Circle the side with fewer presses.

Problem 26 (continued): Try two more pairs. Begin all OFF each time. Press each thick line once and color the lamps that finish ON. Count the presses.

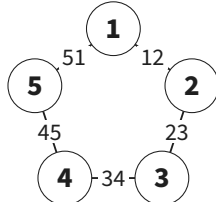


Press these

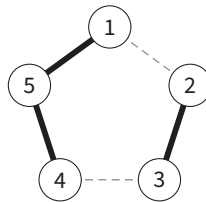


___ presses

Color result

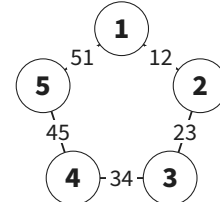


Reset. Press these

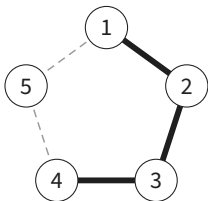


___ presses

Color result

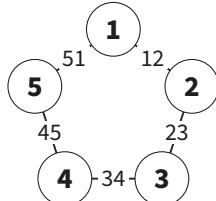


Press these

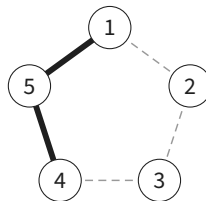


___ presses

Color result

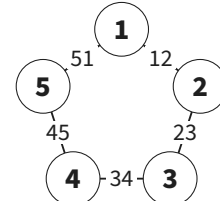


Reset. Press these



___ presses

Color result

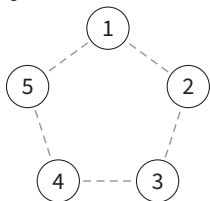


Compare the two final pictures in each pair. Circle the side with fewer presses.

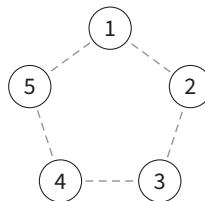
Finish this sentence with words or a drawing: “When I press the other lines, ...”

Problem 27: Make your own pair of line pictures. Choose one, two, or three lines on the left ring and trace them thickly. On the right ring, trace every line you did not choose.

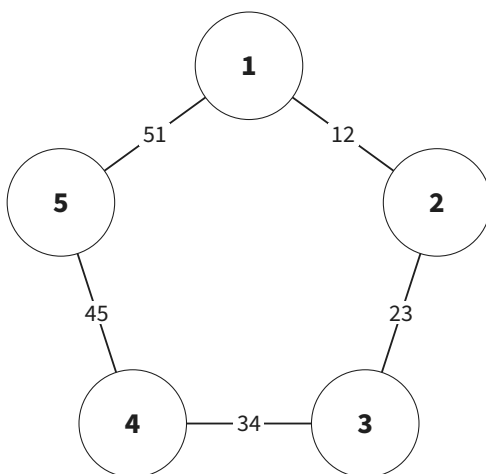
My chosen lines



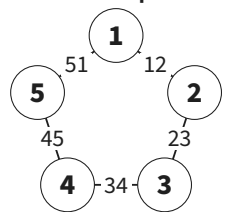
The other lines



Start all OFF on the large mat. Press each chosen line once. Color the first final picture below. Reset and test the other lines; color the second final picture.

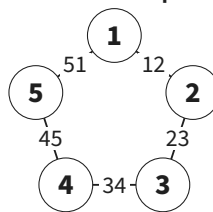


First final picture



_____ presses

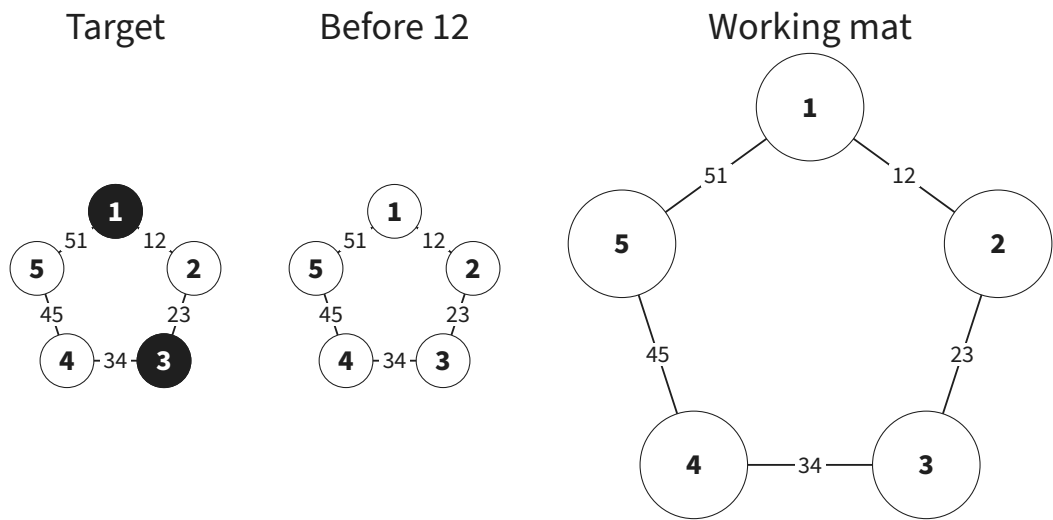
Second final picture



_____ presses

Compare your pictures. Show your partner the pair and ask them to test it.

Problem 28: Make the target below by fixing one lamp at a time. Start all OFF. This time, do not press line 51. Use each line at most once.



Lamp 1 is OFF, but the target needs it ON. **Press 12 now.** Circle “yes” in the first row. Lamp 1 now matches the target.

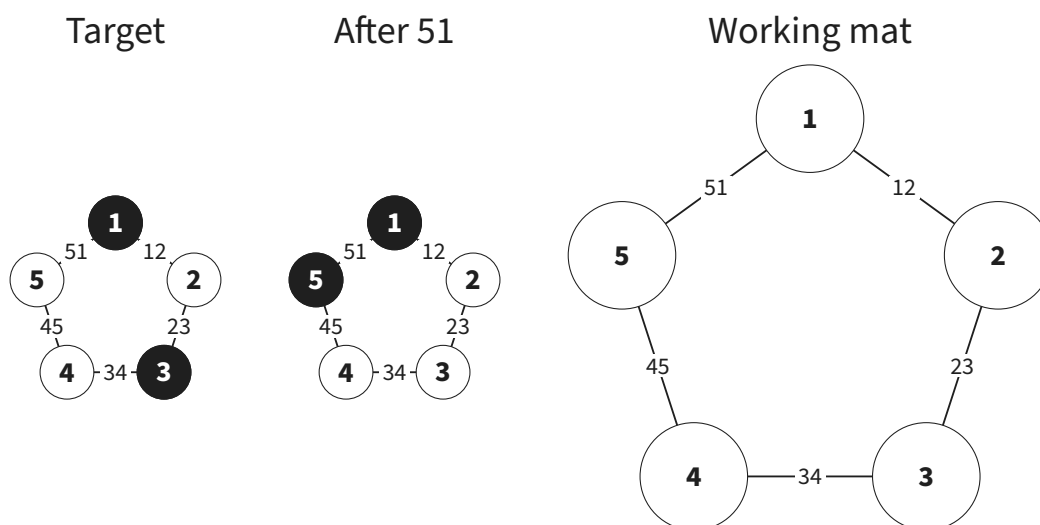
Next look at lamp 2. If it differs from the target, press 23; if it matches, leave 23 alone. Do the same for lamps 3 and 4, using the next line each time.

Look at lamp	To match the target, press this line?	Circle one	
1	12	yes	no
2	23	yes	no
3	34	yes	no
4	45	yes	no

Stop after 45. Does lamp 5 match the target? ☐ yes ☐ no

Write the lines you pressed, in order.

Problem 28 (continued): Try the same target again. Start all OFF. This time, press line 51 first. Then decide about 12, 23, 34, 45, in that order. Use each line at most once.



After pressing 51, lamp 1 is already ON, just as the target needs. **Do not press 12.** Circle “no” in the first row.

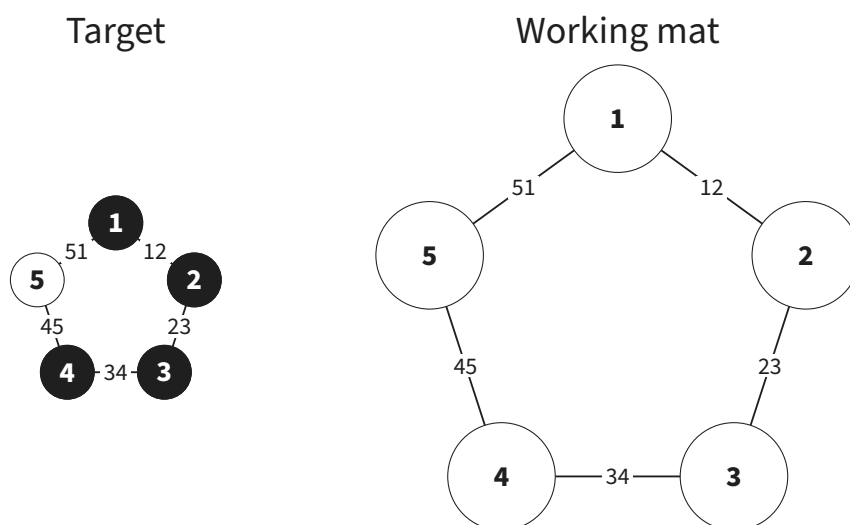
For each next lamp, compare it to the target. Press the next line only if that lamp needs to change.

Look at lamp	To match the target, press this line?	Circle one	
1	12	yes	no
2	23	yes	no
3	34	yes	no
4	45	yes	no

Stop after 45. Does lamp 5 match the target? ☐ yes ☐ no

Write your whole move list, starting with 51.

Problem 29: Make this new target in two ways. Start all OFF for each trial. Decide about 51 first, then 12, 23, 34, 45. Press each line at most once.



Look at each lamp in order. If it differs from the target, press the line below it. If it matches, do not press that line. After lamp 4, stop and check lamp 5. Include 51 in your press count if you used it.

First trial: leave 51 alone.

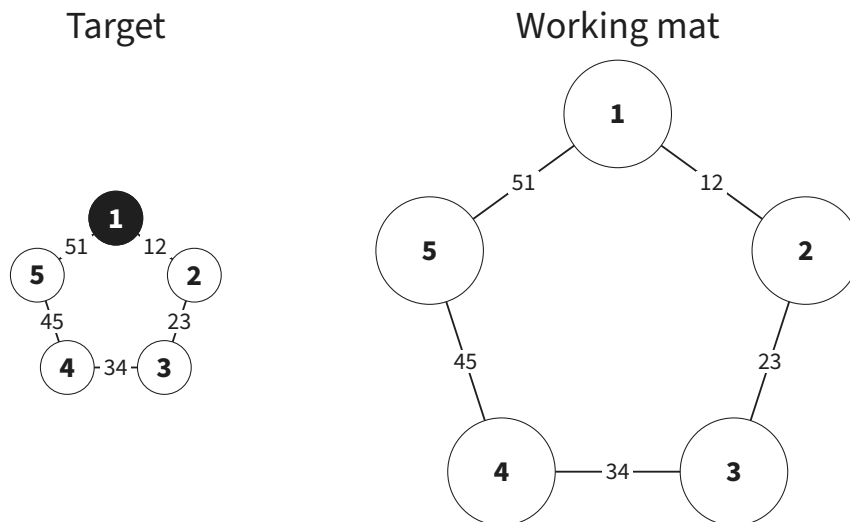
Lamp to fix	1	2	3	4
Line to press?	12	23	34	45
Circle one	yes / no	yes / no	yes / no	yes / no
Lamp 5 matches the target: ☐ yes ☐ no				Total presses: _____

Reset to all OFF. Second trial: press 51 first.

Lamp to fix	1	2	3	4
Line to press?	12	23	34	45
Circle one	yes / no	yes / no	yes / no	yes / no
Lamp 5 matches the target: ☐ yes ☐ no				Total presses: _____

Circle the trial that uses fewer presses. Point to its lines on the working mat.

Problem 30: Try this target: only lamp 1 ON. Start all OFF for each trial. Choose about 51 first, then decide about 12, 23, 34, 45. Press each line at most once.



At each lamp, press the next line only if that lamp needs to change to match the target. After deciding about 45, stop. Check lamp 5 carefully. Include 51 in your press count if you used it.

First trial: leave 51 alone.

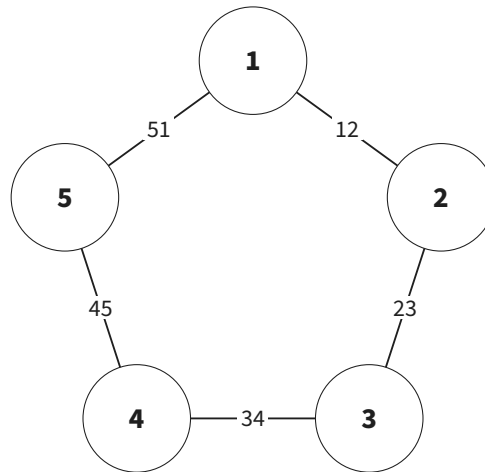
Lamp to fix	1	2	3	4
Line to press?	12	23	34	45
Circle one	yes / no	yes / no	yes / no	yes / no
Lamp 5 matches the target: ☐ yes ☐ no				Total presses: _____

Reset to all OFF. Second trial: press 51 first.

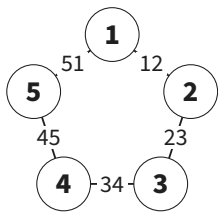
Lamp to fix	1	2	3	4
Line to press?	12	23	34	45
Circle one	yes / no	yes / no	yes / no	yes / no
Lamp 5 matches the target: ☐ yes ☐ no				Total presses: _____

Put a cross beside a trial if its final picture does not match the target. Show a partner which lamp does not match.

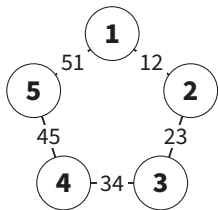
Problem 31: Invent three new targets. In each small ring, color two or four lamps ON. Test each target on the large mat, starting all OFF each time.



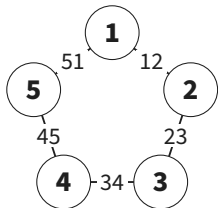
Find one move list using each line at most once. For the other list, use every line missing from the first list, once each. Test both. Count the presses and circle the shorter list.



My target. Counts: _____
 First list: _____
 Another list: _____



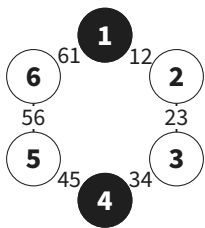
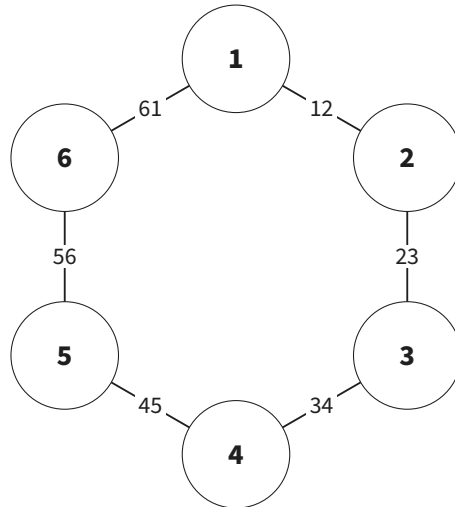
My target. Counts: _____
 First list: _____
 Another list: _____



My target. Counts: _____
 First list: _____
 Another list: _____

If you want another puzzle: can you draw a target with two or four ON lamps that needs more than two presses?

Problem 32: Start all OFF on this six-lamp ring. Follow one route between the dark lamps of a target. Press each line on that route once. Reset and try the other route. Make both targets below.

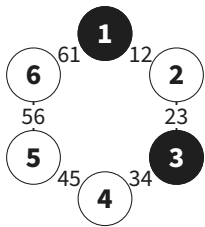


First route: _____

Press count: _____

Other route: _____

Press count: _____



First route: _____

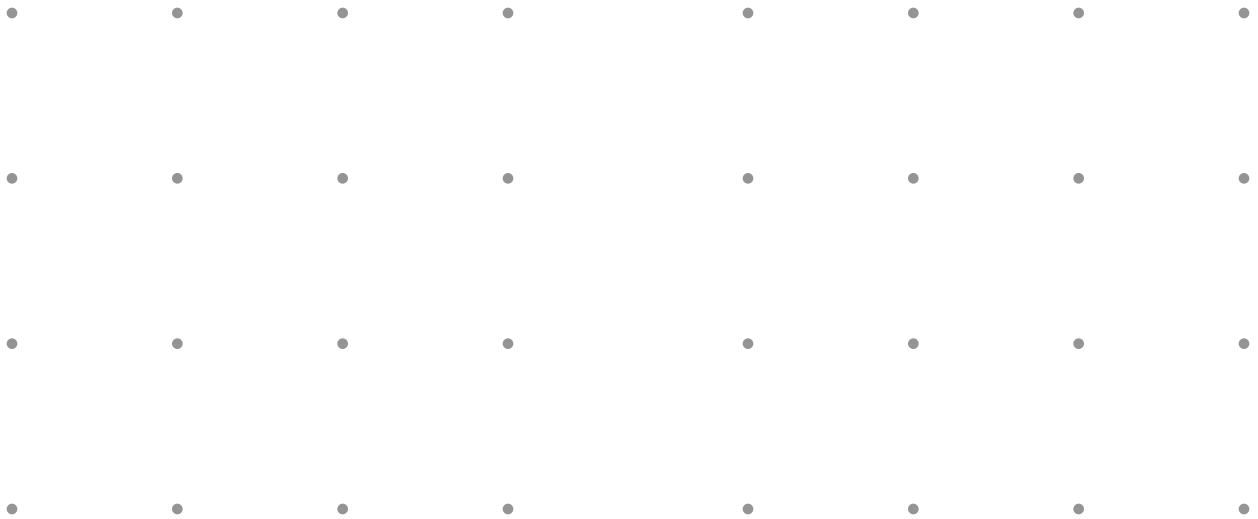
Press count: _____

Other route: _____

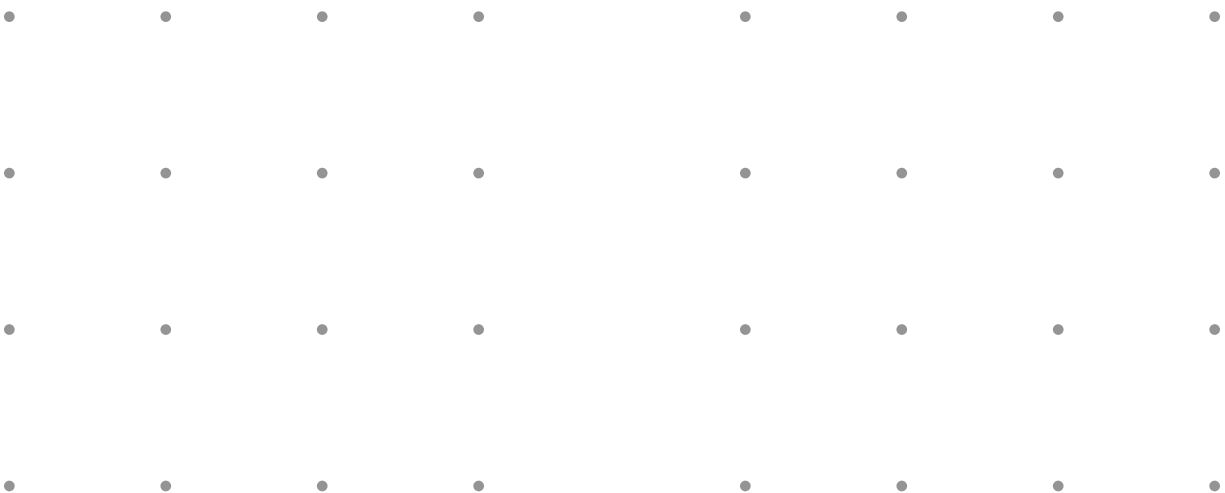
Press count: _____

Circle each pair of routes that uses the same number of presses. Could two routes around the five-lamp ring use the same number? Draw or write your idea.

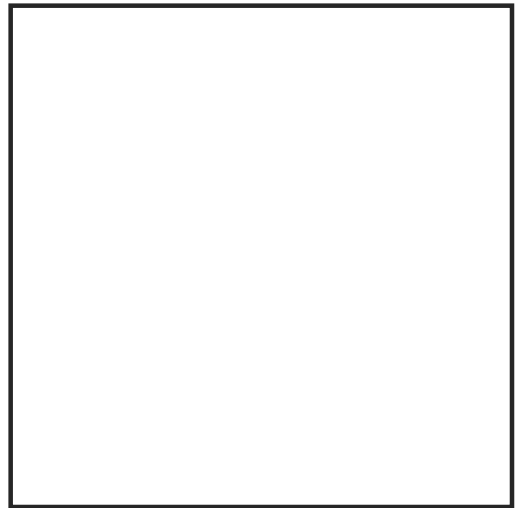
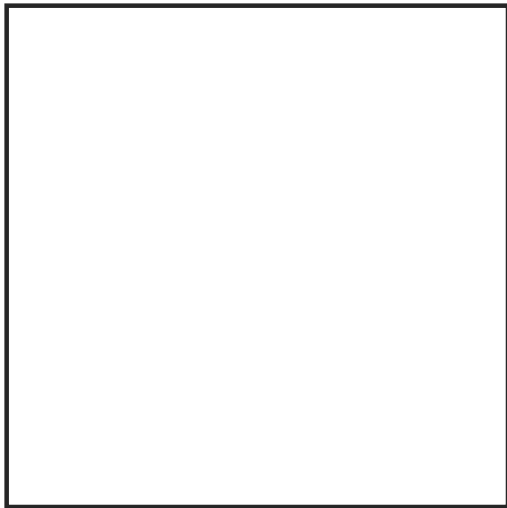
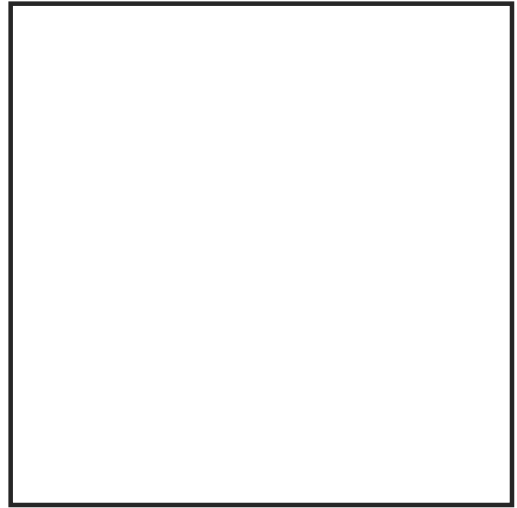
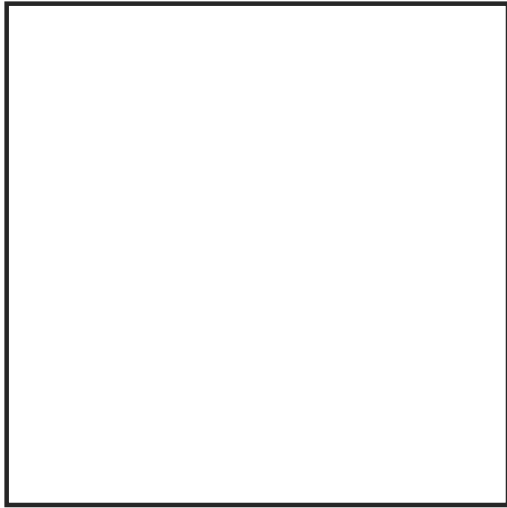
Problem 33: Join dots with straight lines to make a closed shape with THREE corners. Keep your lines from crossing. Make two different shapes.



Problem 34: Now make two different closed shapes with FOUR corners. Touch each corner where your line turns.

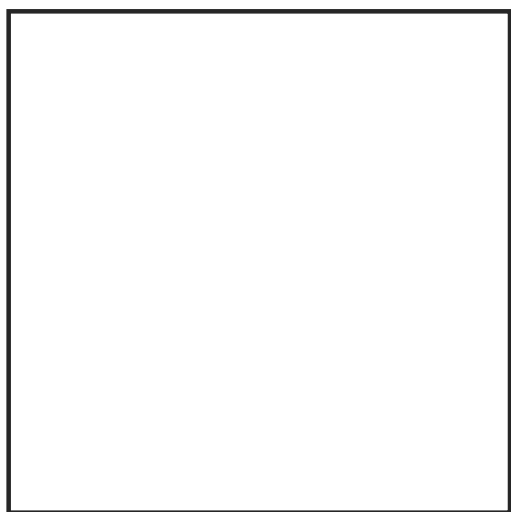
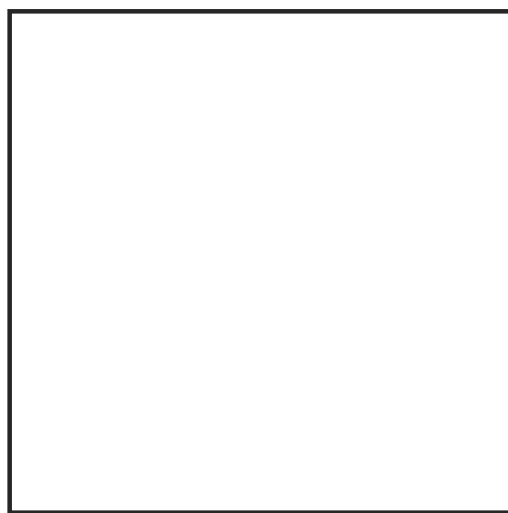
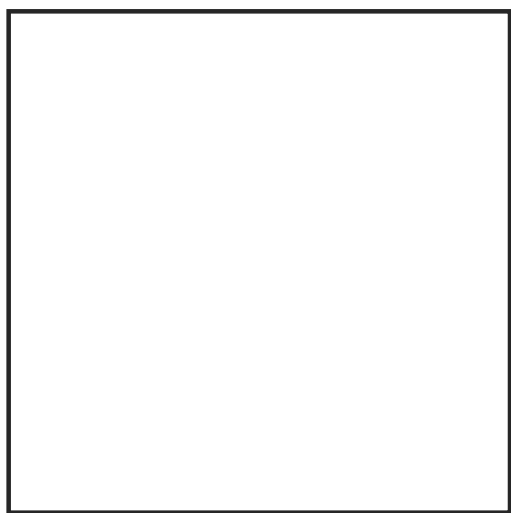


Problem 35: Each square is a house. Draw ONE straight wall all the way across each house, from its outside edge to its outside edge. Make two rooms. Try different walls.

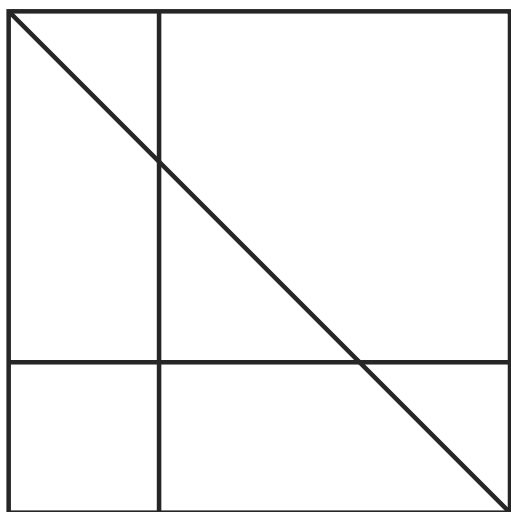
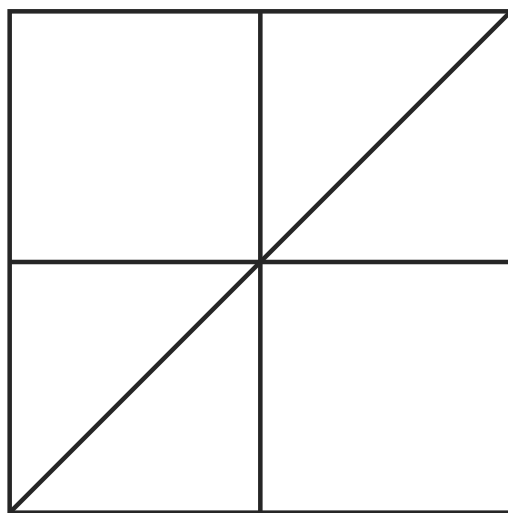
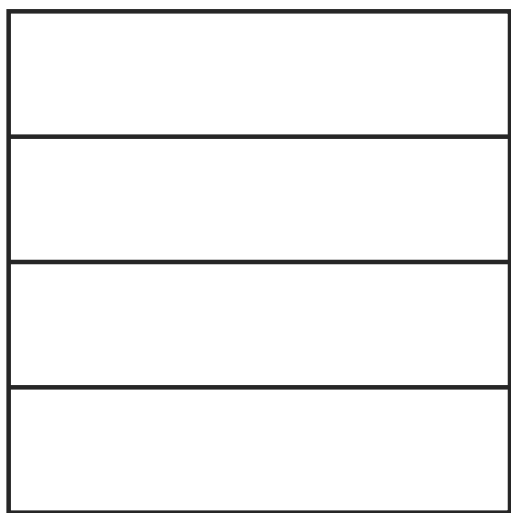


Problem 36: Make or find a house with two triangle rooms. Make or find one with two four-corner rooms. Draw on another sheet if you want to try more houses.

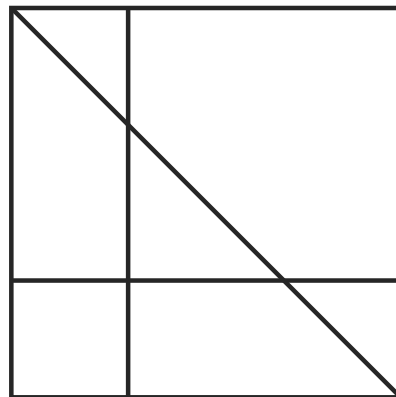
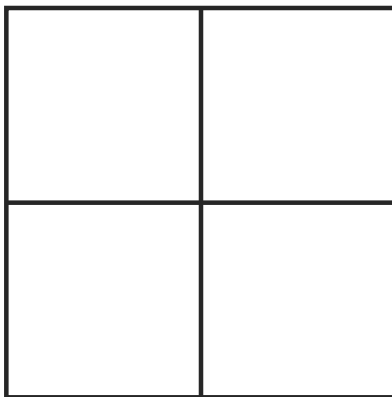
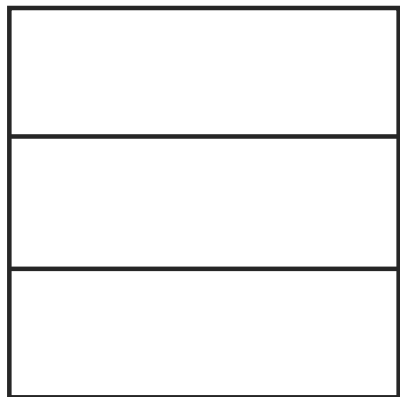
Problem 37: Draw TWO different straight walls all the way across each house. Walls may cross. Put one dot in each room. Try to make different numbers of rooms.



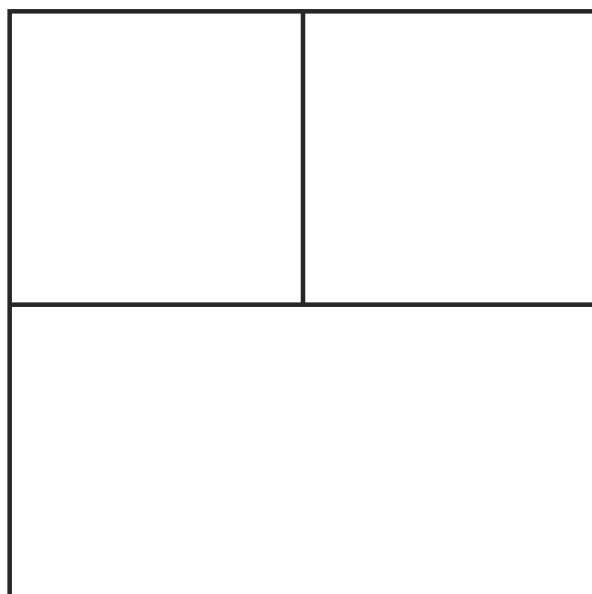
Problem 38: These houses each have THREE straight walls. Put one dot in every room. Which has the most rooms? In the empty house, try three different walls of your own.



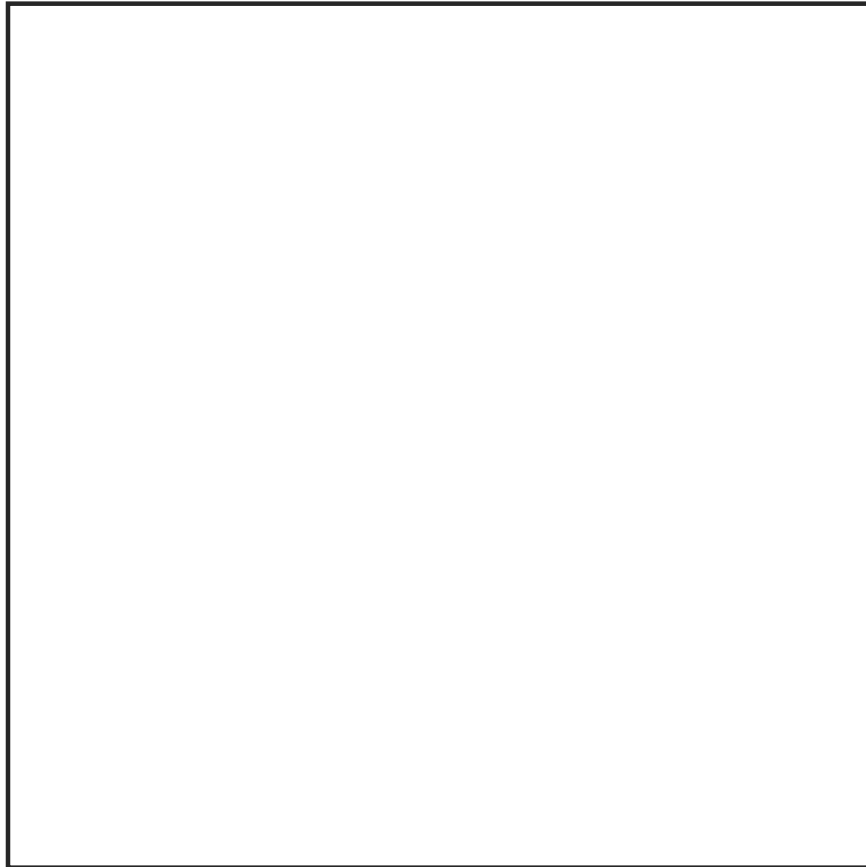
Problem 39: Put a dot or an X in every room in the first three houses. Rooms sharing a wall must have DIFFERENT marks. Rooms touching only at a corner may match.



Problem 40: Try this house. If two marks do not work, you may also use an O.

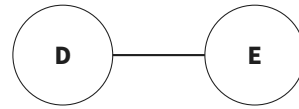
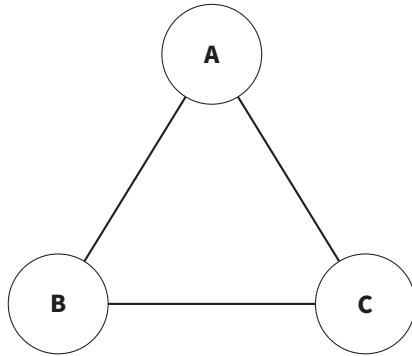


Problem 41: Make a house with one, two, or three different straight walls. Draw each wall all the way across. Let a partner put one dot in every room. Swap jobs.

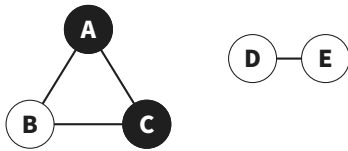


Problem 42: Make a giant house together on a whiteboard or another sheet. Take turns adding one new straight wall all the way across. Do not trace an old wall. Stop after three walls. Find every room together.

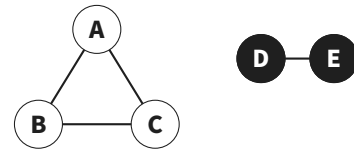
Problem 43: Start all OFF on the large mat. Press a line to flip the lamps at both ends. Make each target below: dark means ON; white means OFF. Use dots you can erase or counters you can flip.



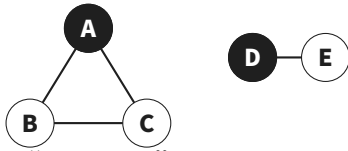
Write **AC** for a press of the line joining A and C. A move cannot jump across the gap. Reset to all OFF for every target.



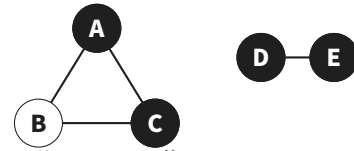
Moves, or “not yet”:



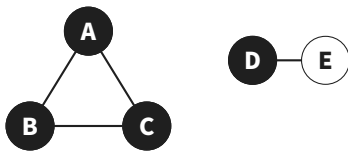
Moves, or “not yet”:



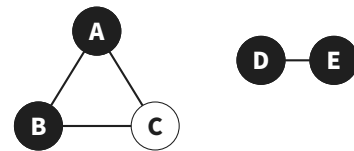
Moves, or “not yet”:



Moves, or “not yet”:

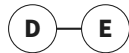
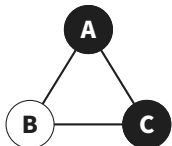
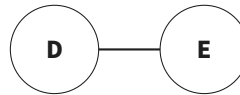
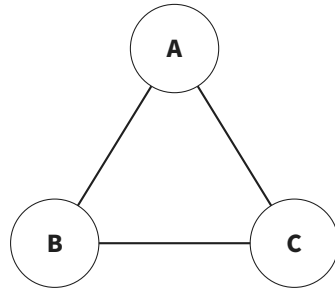


Moves, or “not yet”:

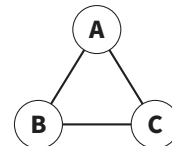


Moves, or “not yet”:

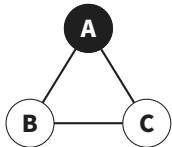
Problem 44: Try these targets, starting all OFF each time. In each picture, count the dark lamps in the triangle. Then count the dark lamps in the pair. Write the two counts and circle “made” or “not yet.”



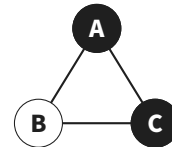
ON in triangle: ____ in pair: ____
Made / not yet



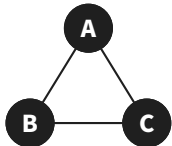
ON in triangle: ____ in pair: ____
Made / not yet



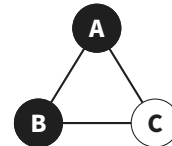
ON in triangle: ____ in pair: ____
Made / not yet



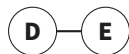
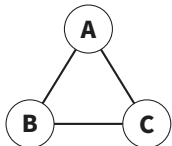
ON in triangle: ____ in pair: ____
Made / not yet



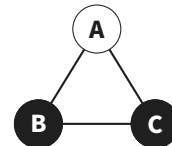
ON in triangle: ____ in pair: ____
Made / not yet



ON in triangle: ____ in pair: ____
Made / not yet



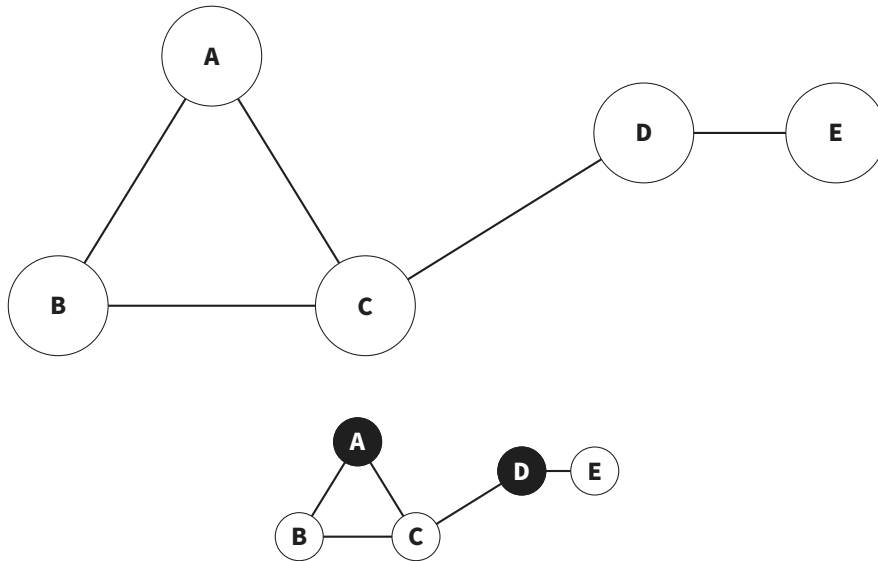
ON in triangle: ____ in pair: ____
Made / not yet



ON in triangle: ____ in pair: ____
Made / not yet

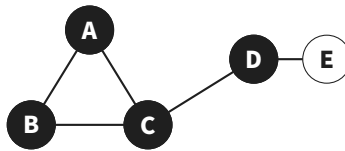
Point to the pictures you made. What do you notice about their two counts? Show a partner a rule that predicts which pictures can be made.

Problem 45: The two pieces now have a bridge: line CD. Start all OFF and try each target below. Press a line to flip both ends. Write the lines you press in order.



Make this picture. Moves: _____

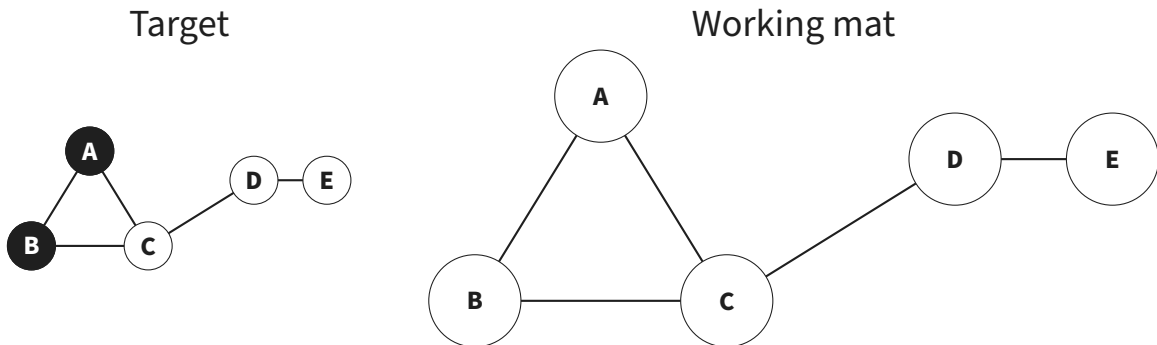
Reset to all OFF. Make the next picture.



Moves: _____

Point to a line your solution needs that was missing from Problem 43. What changed when we added it? Tell a partner, or draw your idea.

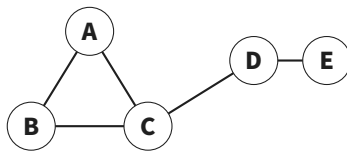
Problem 46: Make the pictured target by trying the two routes shown below. Begin all OFF on the large mat. Press the lines in the first route, then the lines in the second route.



First route: A–C–D. Then: B–C–D.

Try the full list: **AC, CD, BC, CD.**

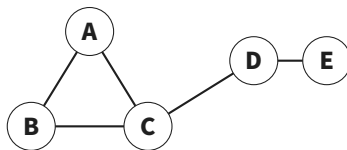
Color the lamps that finish ON in this small picture.



In the list below, cross out both copies of the repeated press.

AC CD BC CD

Reset to all OFF. Try only the presses you did not cross out. Color that final picture below.



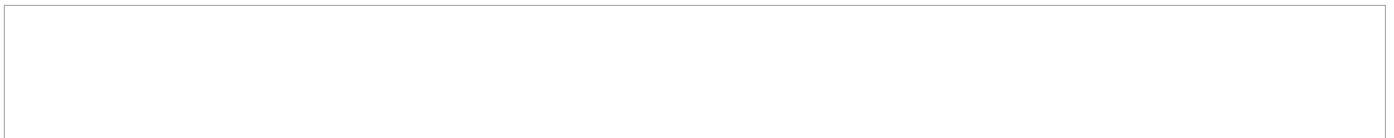
Compare your two pictures with the target. Do both lists work?

Find a longer list of your own that makes this target. Give it to a partner to shorten and test.

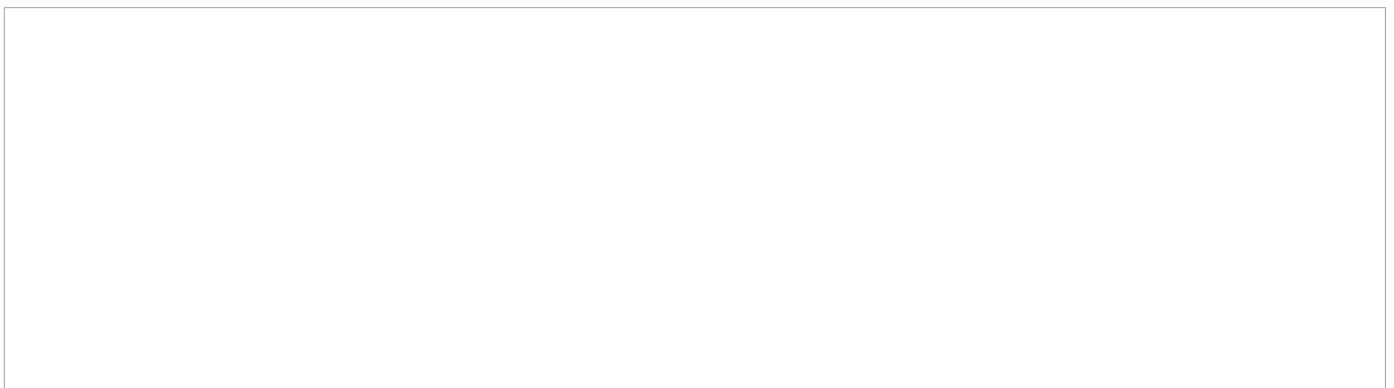
Problem 47: Invent a network with two separate pieces. Draw at least three lamps in each piece, joined by lines. Leave a gap with no line between the pieces.



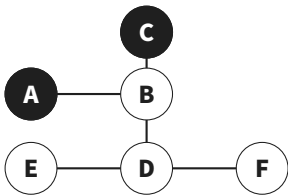
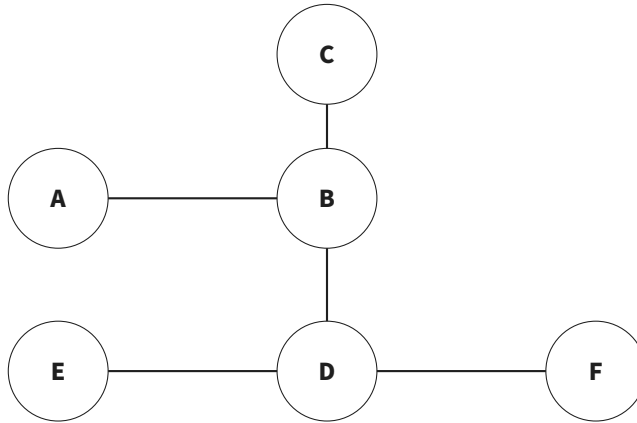
Choose one target you predict can be made from all OFF. Show it by coloring lamps in your drawing. Try it with dots on a whiteboard, or counters on this page. Write a move list or trace the lines you used.



Copy your network below. Color a target you predict cannot be made from all OFF. Ask a partner to try it. Point to the lamps that helped you choose it.



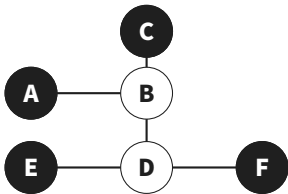
Problem 48: This network has no closed loop. Start all OFF. Press a line to flip both ends. Make each pictured target using every line at most once. Use dots you can erase or counters you can flip.



Make this picture.

Moves: _____

Reset to all OFF before making the next picture.

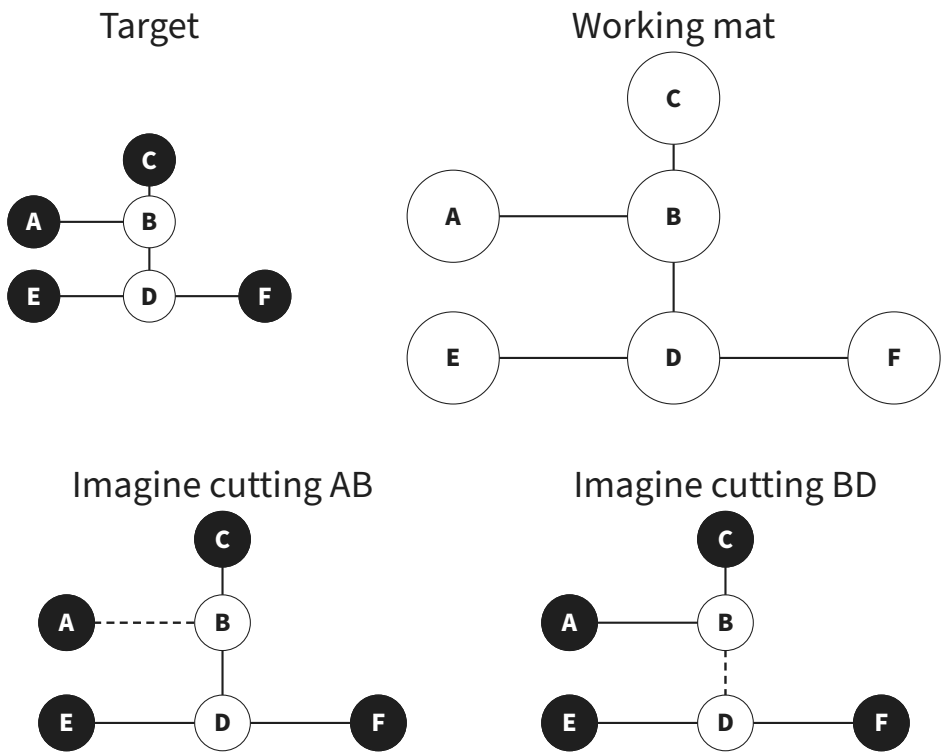


Make this picture.

Moves: _____

Choose one target. Trade move lists with a partner and replay their list. Point to any lines you both used.

Problem 49: Use the target shown below. Imagine cutting one line at a time. Count the dark target lamps on the named side of the cut. Predict which lines to press, then test from all OFF.



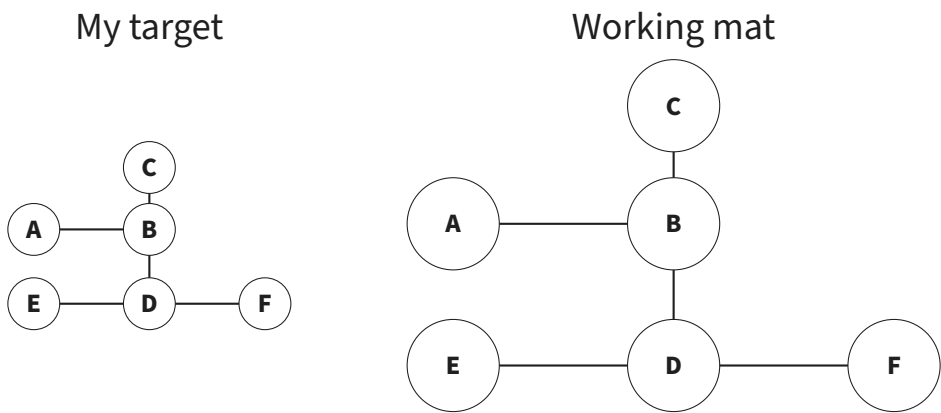
At cut AB, the A side has **1** target lamp. Press AB once. At cut BD, the A,B,C side has **2** target lamps. Leave BD alone.

Complete the other rows. Use each line at most once.

Cut line	Count on this side	Target count	Press?
AB	A	1	yes
BC	C	—	yes / no
BD	A, B, C	2	no
DE	E	—	yes / no
DF	F	—	yes / no

Try your predicted list. Does it make the target? Show a partner how the counts helped you choose the lines.

Problem 50: Invent another target on this tree. Color 2, 4, or 6 lamps ON in the small picture. Choose a different target from Problem 48. Leave the large working mat all OFF.



Before trying moves, imagine cutting each line. Count the ON lamps in your target on the named side. Predict whether to press the line. Use each line at most once.

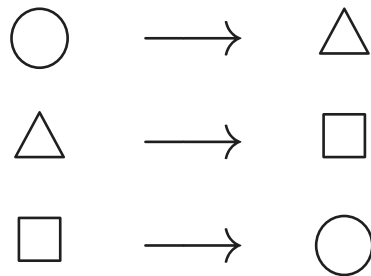
Cut line	Count on this side	Target count	Press?
AB	A	_____	yes / no
BC	C	_____	yes / no
BD	A, B, C	_____	yes / no
DE	E	_____	yes / no
DF	F	_____	yes / no

Write your predicted move list: _____

Now test it from all OFF. Compare the result with your target. If they differ, circle a row to check again.

Trade targets with a partner. Predict the presses for their target and test them together.

Problem 1: One turn changes each symbol using this key. Keep its place in the message. Point to an input and follow its arrow. Try each symbol.



Problem 2: Send each message through the key. Draw or place its new symbols in the space beside it.

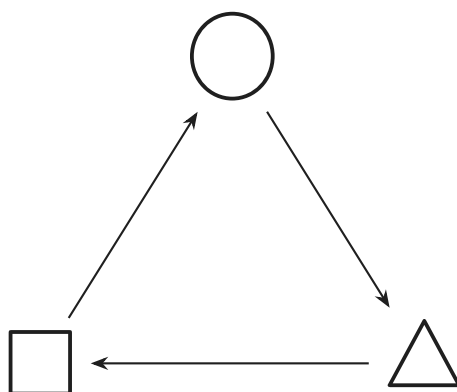
Message	New message
○ △ ○	
□ □ △	
△ ○ □	

Problem 3: A friend received triangle, circle, triangle. Follow the key backward to find the original message. Check by sending it forward again.

Problem 4: Use the same key. Start with a circle card. After every turn draw what it becomes. Keep going until the circle first comes back. Repeat from triangle and from square.

Start	Turn 1	Turn 2	Turn 3
○			
△			
□			

Problem 5: These arrows show the same key as a loop. Move a counter one arrow each turn. Try all three starting shapes. Mark each arrival at its starting shape.



Problem 6: Start with circle, triangle, square in a row. Change all three each turn. Stop when the whole row is back. Draw the row just before it comes back.

Problem 7: Change the key: circle and triangle swap; square stays square. Try all three starting shapes. Draw their results for two turns.

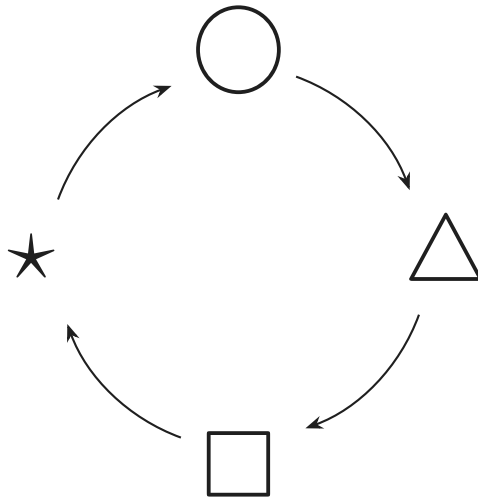


Start	Turn 1	Turn 2
		
		
		

Problem 8: Start with all three shapes in a row. Repeat this new key until the whole row returns. Compare with the old key. Record the number of turns for each.

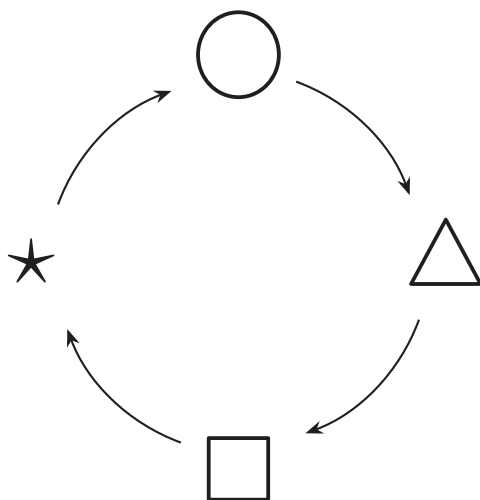
Problem 9: Try another key: circle becomes square, and triangle also becomes square. Secretly choose circle or triangle and show only its output. Can your partner tell which input you chose? Swap roles and try again. Draw the two possible inputs.

Problem 1: Put a counter on circle. One hop follows an arrow to the next shape. Move one hop at a time until you return to circle. Draw each place you land, ending with circle.



Problem 2: Start at triangle. Take one hop each turn until you return. Draw the landings. Compare with your route from circle.

Problem 3: Start at circle. Now take two hops every turn. Draw only the shape where each turn ends. Stop when circle returns. Repeat from triangle.



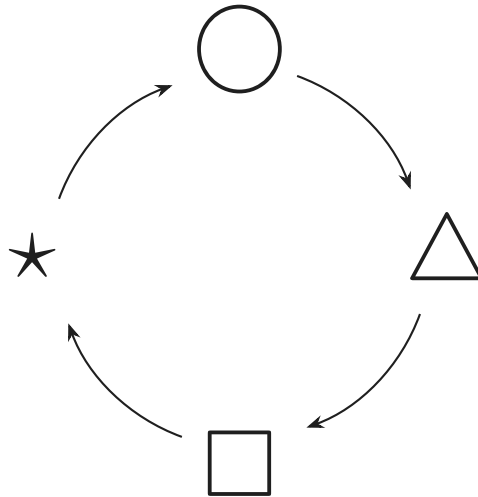
From circle:

From triangle:

Problem 4: Try to land on triangle by taking two hops each turn from circle. Then change your starting shape to square and try again. Circle the shapes you can reach from these two starts.



Problem 5: Start at circle. Take three hops every turn. Draw each landing until circle returns. Compare the shapes visited by one-hop, two-hop, and three-hop turns.



Problem 6: A secret turn moves every counter the same number of clockwise hops, fewer than four. Circle becomes square. Use the ring to find where triangle, square, and star go. Draw their outputs.

Problem 7: Try one turn that undoes that secret turn. Test it from each of the four starting shapes. Show someone each out-and-back trip.

Problem 1: Use towers of heights 1, 2, and 3. Put them in the order 2, 1, 3 from left to right. Look from the left, then the right. A taller tower hides a shorter one behind it. Write 2, 1, 3 from left to right in both rows below. Circle the towers you see from each end.

Looking from the left

left

right

Looking from the right

left

right

Problem 2: Rearrange the towers into 1, 3, 2. Look from both ends again. Draw or write the tower heights. Circle the towers you see.

Looking from the left

left

right

Looking from the right

left

right

Problem 3: Rearrange the same three towers for each view below. Draw or write each row from left to right.

Towers seen from the left	My tower order
3	
1	
2	

Problem 4: Build different rows with heights 1, 2, and 3. Keep the left end in the same place. Record each new order and count the towers seen from each end. Do not record the same order twice.

Row	Tower order, left to right	Seen left	Seen right
1			
2			
3			
4			
5			
6			

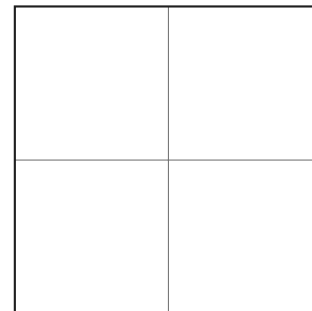
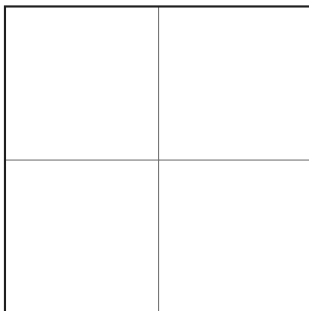
Problem 5: Compare your recorded rows. Circle the rows that show two towers from the left and two from the right. Build each circled row again and check both views.

A row I checked:

Problem 6: Try to make a row of heights 1, 2, and 3 that shows only one tower from each end. Record two different attempts. For each attempt, count from both ends.

Tower order	Seen left	Seen right

Problem 7: Make a tiny city using two towers of height 1 and two of height 2. Put one tower in each square. Every row and every column must contain both heights. Record two different cities. Circle the city whose top row shows two towers from the left.



Problem 1: Use R and B counters; repeats are allowed. A test scores one point for each matching position. Say only the total score. Keep the secret RB visible. Build each test below. Point to matching positions, then record the total.

Test	Score against secret RB
RR	
RB	
BR	
BB	

Problem 2: Hide a two-place secret. Your partner makes tests and you give scores. Keep testing until your partner can name the secret. Record tests below or on scrap paper. Reveal the secret and check the scores. Swap jobs and play again.

First secret: _____

Next secret: _____

Test	Score

Test	Score

Problem 3: Build the four possible secrets RR, RB, BR, BB. Test RR against each one and record its score. Circle every secret that could remain if the answer were 1.

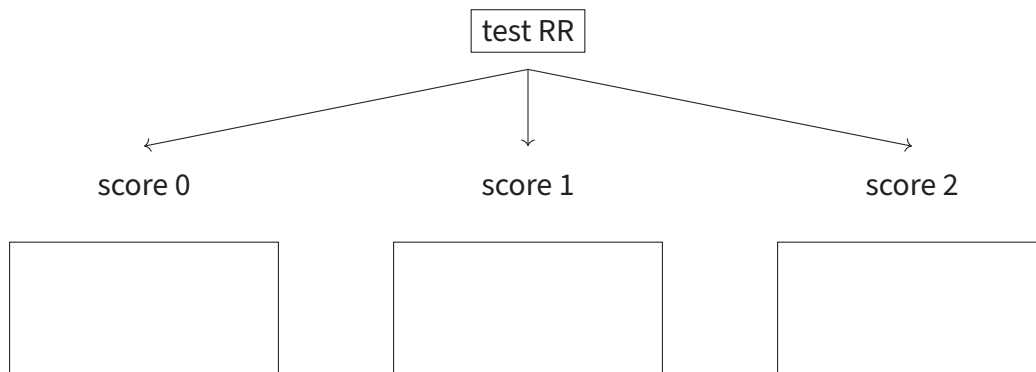
Possible secret	Score for test RR
RR	
RB	
BR	
BB	

Problem 4: Keep just the two circled secrets. Try RR again, then try RB. Record the score each test would receive for each remaining secret. Circle a test that tells the secrets apart.

Test	Score if secret is RB	Score if secret is BR
RR		
RB		

A different test that also tells them apart:

Problem 5: Begin with test RR. Under each answer, write or build every secret that could remain. Use your checks from Problem 3.



Problem 6: If a branch has two possible secrets, choose a test and record its score for each secret. If a branch has one secret, name it without another test.

My next test and what each answer means:

Problem 7: Ask your partner to hide each of RR, RB, BR, and BB in an order you do not know. Follow your branches. Record how many tests you needed before you could name each secret.

Secret after reveal	RR	RB	BR	BB
Number of tests				

Problem 1: Two players alternate taking 1 or 2 counters. Never take more than remain. Taking the last counter wins. Play from 4 counters twice, swapping who starts. Draw or write the number taken at each turn. Circle the last move.

Game 1 moves:

--

Game 2 moves:

--

Problem 2: Play from 5 counters twice, swapping who starts. Record the moves and circle the last move. Use the same 1-or-2 rule.

Game 1 moves:

--

Game 2 moves:

--

Problem 3: Two players alternate taking 1 or 2 counters. Never take more than remain. Taking the last counter wins. Start with 3 counters. Let the other player start. Try each of their first moves below. Record how many you take next.

Starting pile	Other player takes	I take next	Counters left
3	1		
3	2		

Problem 4: Start with 6 counters. Make two groups of 3. Let the other player start. Try each first move below and choose your reply so that exactly 3 counters remain. Then finish each game.

Other player takes	I take next	Counters left
1		
2		

Draw the two groups before the game:

Problem 5: Two players alternate taking 1 or 2 counters. Never take more than remain. Taking the last counter wins. Try 7, 8, and 9 counters. For each pile, choose whether to go first or second. If you go first, record your first move. Play twice from each pile and check your choice.

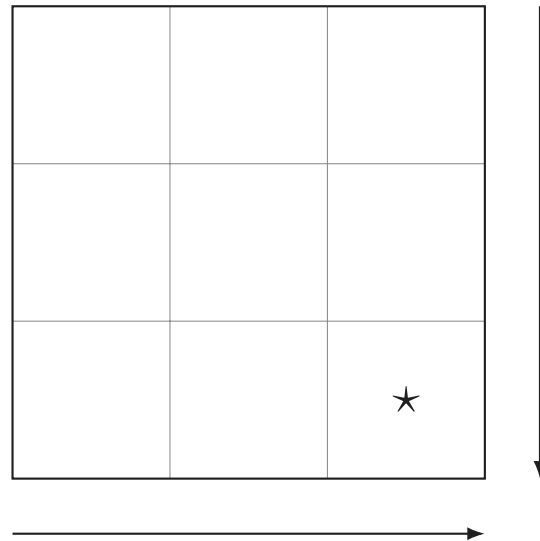
Pile	I choose first or second	My first move if I start
7		
8		
9		

Problem 6: Build 12 counters as four groups of 3. Let the other player start. After each of their moves, take enough to remove 3 counters between your two turns. Record the numbers left after each of your turns. Play until the counters are gone.

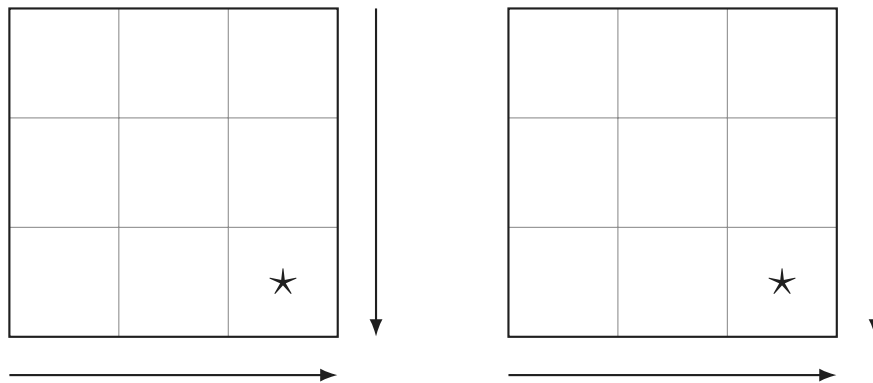
Draw my four groups:

Counters left after my turns:

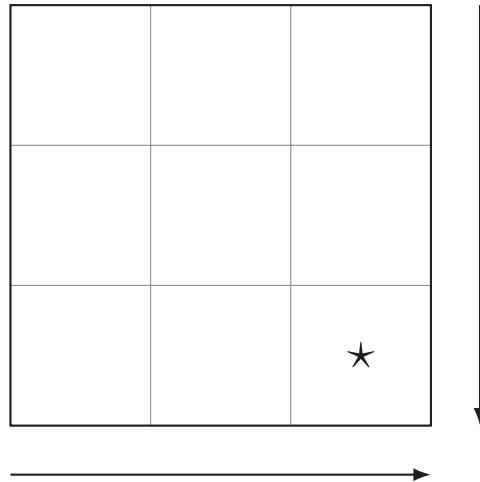
Problem 1: Share one token. Take turns moving it right or down. Move one or more squares; do not move diagonally. Whoever reaches the star wins. Play from the square just above the star, then from the middle square. Switch who starts and play both again.



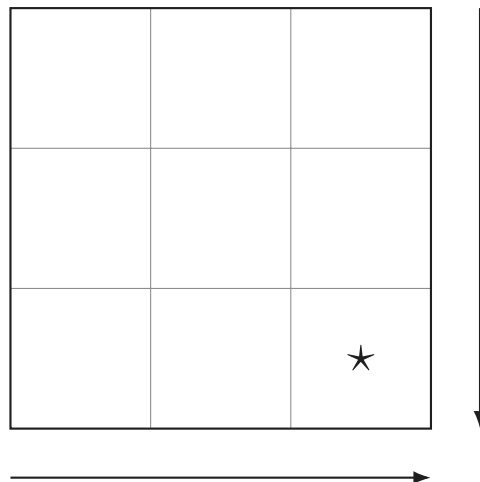
Problem 2: Put the token in the middle. Draw each possible first move on a small board. Then draw the next player's move to the star.



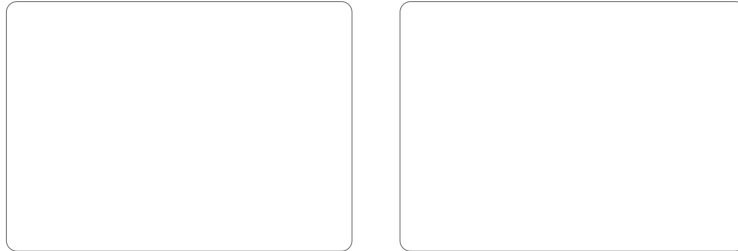
Problem 3: Start in the top-left square. Play twice, switching who starts. Try a long first move in one game and a short first move in the other. Draw a first move and its reply below.



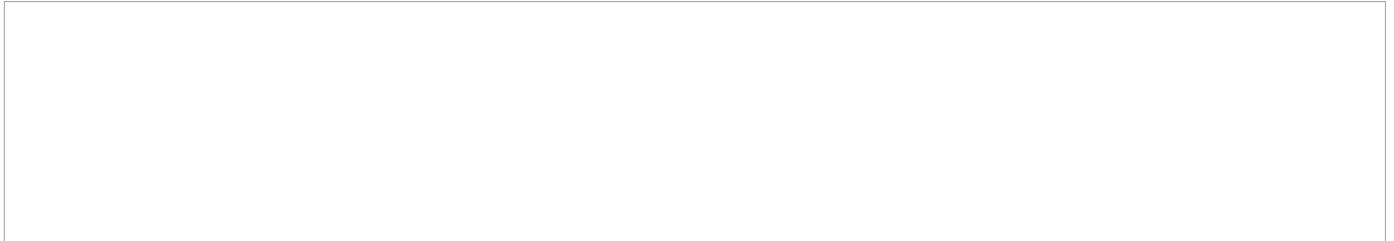
Problem 4: Now start in the left square of the middle row. Play twice, switching who starts. Choose a first move that leaves the same distance right and down to the star. Draw that move.



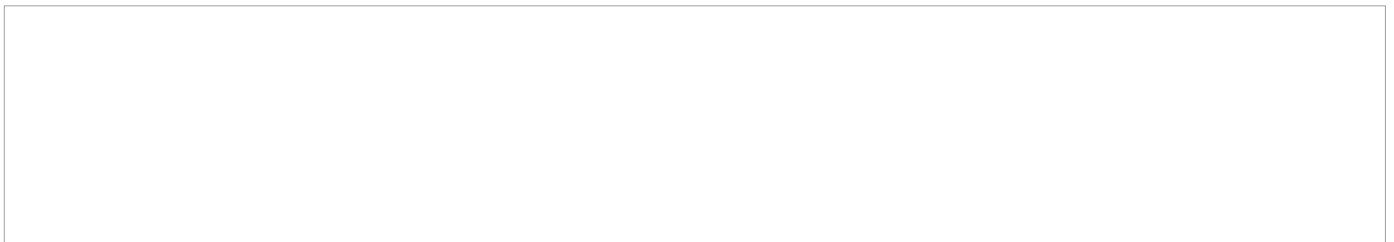
Problem 5: Build two piles of 2 counters in these spaces. Take turns removing one or more counters from just one pile. Whoever takes the last counter wins. Play twice, switching who starts.

Two empty rectangular boxes, each intended for a pile of 2 counters.

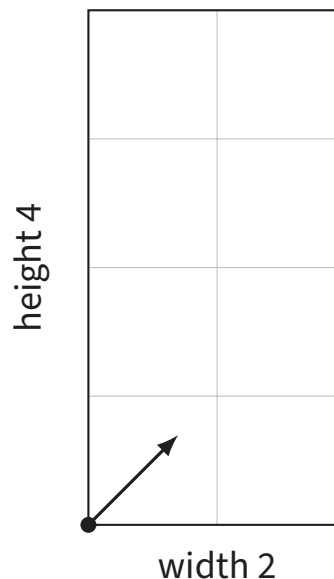
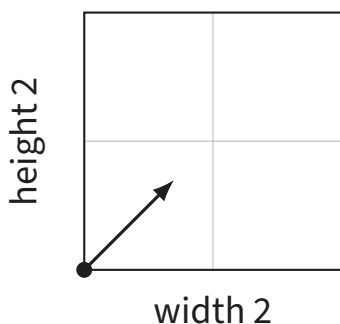
Problem 6: Build two piles of 3. After your partner takes counters, take the same number from the other pile. Keep playing that way. Draw the piles after one pair of turns.

A large empty rectangular box for drawing the piles after one pair of turns.

Problem 7: Build piles of 3 and 1. Choose a first move that leaves equal piles. Keep playing. Then choose two new unequal piles and ask a partner to make them equal in one move. Draw your new piles.

A large empty rectangular box for drawing the new piles.

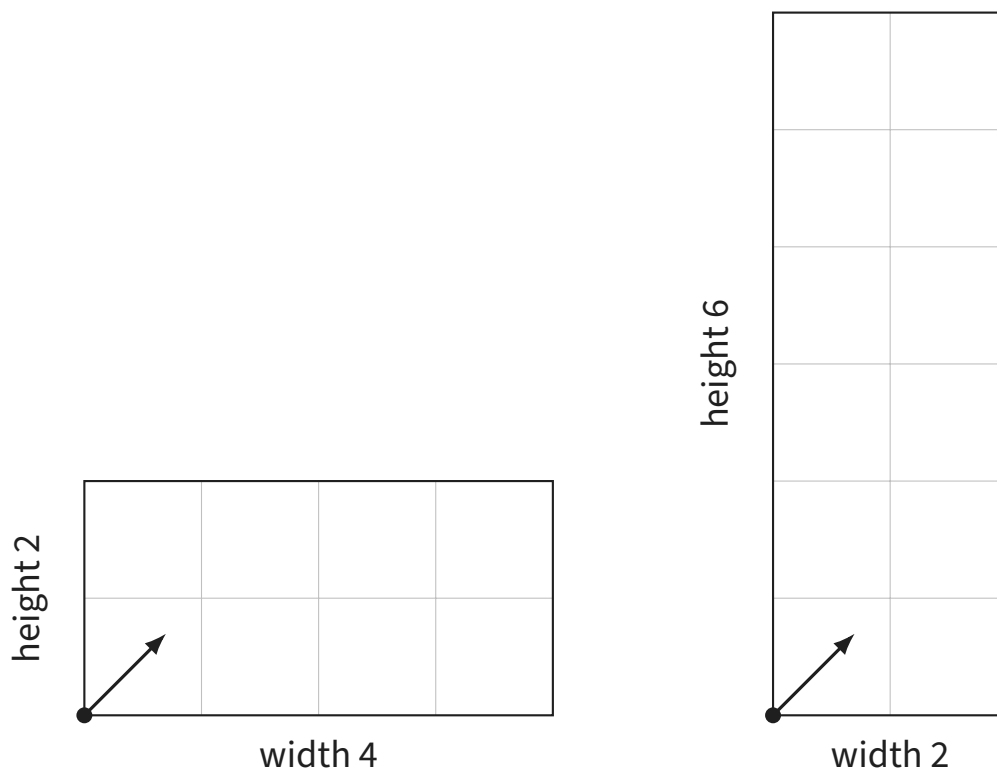
Problem 1: Move a counter from the black dot diagonally up and right. At a side wall turn left or right; at the top or bottom turn down or up. Keep following the diagonals. Stop at the first corner. Try both tables. Draw the paths and circle the ending corners.



Problem 2: Move along the tall table again. At its right wall, show two possible next directions with your finger. Choose the one that stays inside and keeps going up. Draw that turn below.

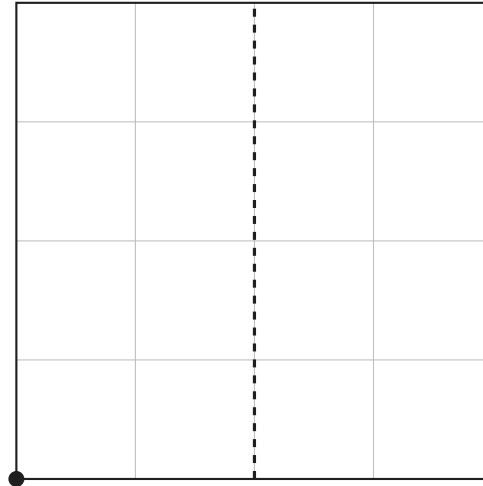
A large empty rectangular box for drawing the path on the tall table.

Problem 3: Predict where each path will end. Put a small dot at your prediction. Trace the path from the black dot, bouncing at walls and stopping at the first corner. Circle the corner you reach.

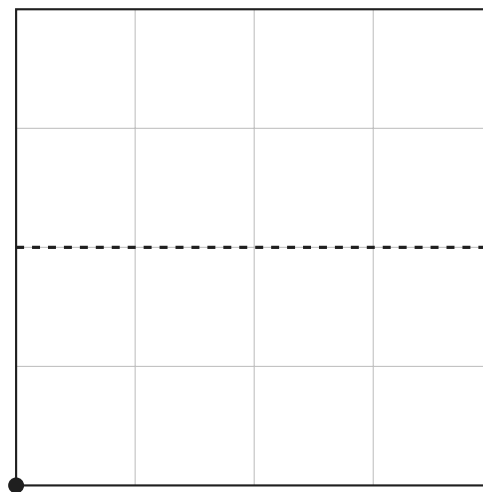


Problem 4: Choose two tables on these pages that finish at different corners. Move a counter along each. Draw the final piece of each path below.

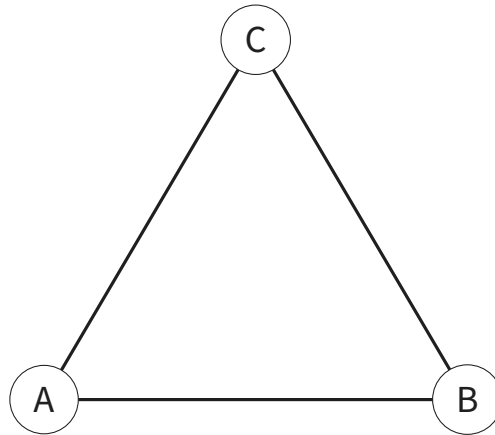
Problem 5: The dashed line below is the wall between two mirror copies of the tall table. Draw one straight diagonal from the black dot to the far corner. Fold on the dashed wall. Trace the two pieces of your line as a bouncing path.



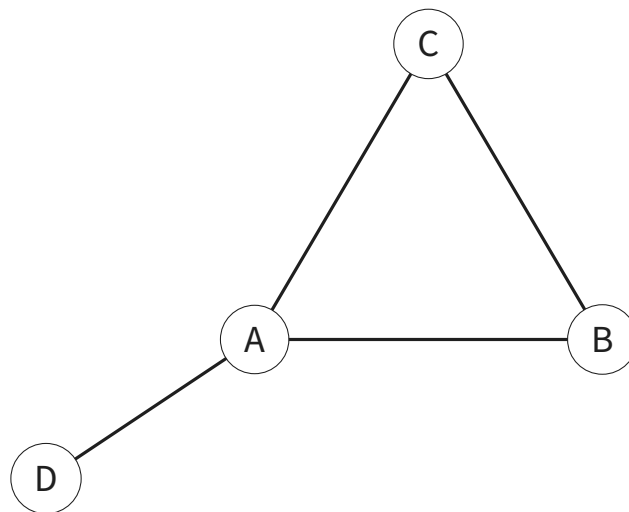
Problem 6: These are mirror copies of the wide table. Draw the same straight diagonal, then fold on this dashed wall. Trace the bouncing path and point to its ending corner. Compare it with Problem 5.



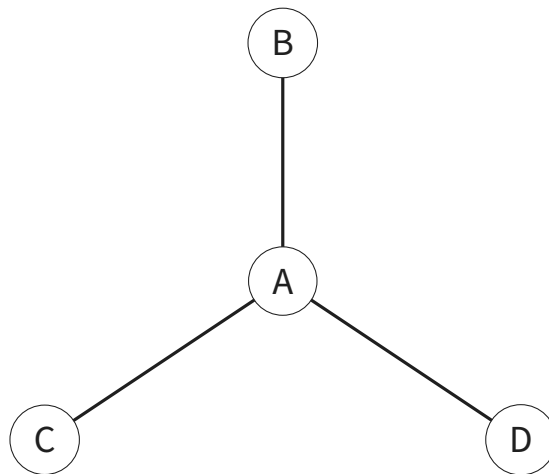
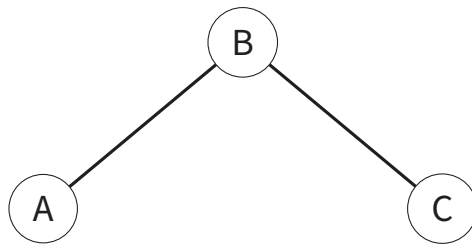
Problem 1: Dots are islands; lines are bridges. Move a counter along every bridge once without jumping. You may visit an island again. Mark a bridge after you use it. Try this town from A, then from B. Draw arrows for one complete walk.



Problem 2: This town has one extra bridge. Try a walk from D, then try from B. Mark used bridges as you go. Circle a place where you can start a walk using every bridge once.

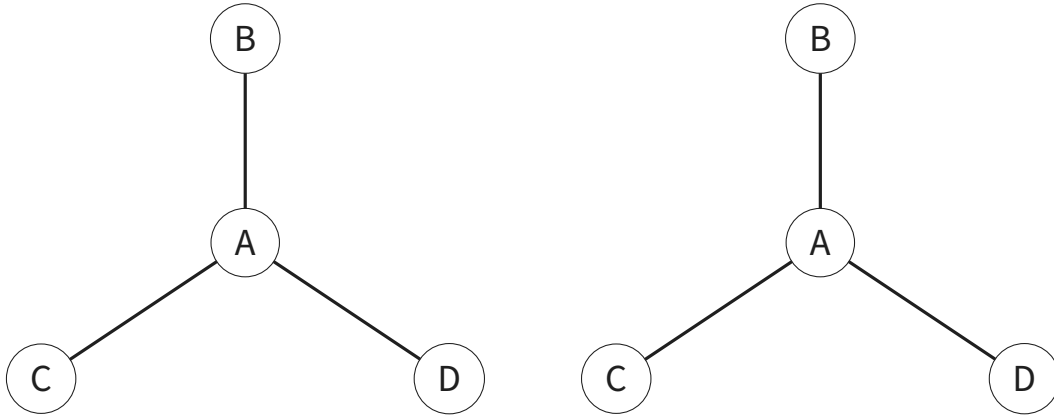


Problem 3: Try a walk using every bridge once in each town. Mark used bridges. Circle where a complete walk starts and ends. If you get stuck, reset and try a different start.



Problem 4: On the three-branch town, try starting at A and then at B. Draw your two attempted walks below. Mark the bridges left unused.

Problem 5: Add one bridge between two outside islands in the left town. Try a walk using every bridge once. Add a different bridge to the right town and try again. Number the bridges in walking order or draw arrows.



Problem 6: Choose one repaired town. Circle its starting and ending islands. Put a small mark on the island with only one bridge. Move the counter along the complete route again.

Problem 7: Draw a small town of your own with three or four islands. Add bridges, then try to use every bridge exactly once. Change a bridge if needed. Show a partner the walk.